

# **AI BASED CONTEXTUAL ANSWER EVALUATION SYSTEM**

## **PHASE 1 REPORT**

Submitted by

**ROSHINI V -230701270**

**SAKSHI P – 230701279**

**SANDHIYA P – 230701282**

**SHRADHA S - 230701311**

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# **ANNA UNIVERSITY, CHENNAI**

## **BONAFIDE CERTIFICATE**

Certified that this Report titled " **AI Based Contextual Answer Evaluation System** " is the bonafide work of **Roshini V (230701270), Sakshi P (230701279), Sandhiya P (230701282), Shradha S (230701311)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

This project addresses the following Sustainable Development Goal:

**Goal- No 4: Quality education**

### **SIGNATURE**

**Dr. J. Manoranjini.,**

Professor and Head,

Department of Computer Science and

Engineering,

Rajalakshmi Engineering College,

Chennai-602 105.

### **SIGNATURE**

**Mrs. S. Punitha,**

Assistant Professor,

Department of Computer Science and Engineering,

Rajalakshmi Engineering College,

Chennai-602 105.

Submitted to Project Viva-Voce Examination held on \_\_\_\_\_

**Internal Examiner**

**External Examiner**

# ABSTRACT

The evaluation of descriptive answers is an important process in the educational system as it helps in assessing a student's understanding and knowledge of a subject. Traditional manual evaluation requires teachers to read and analyze each answer individually, making the process time-consuming and difficult when a large number of students are involved. Existing automated evaluation systems mainly depend on keyword matching and may not effectively understand answers expressed using different words or sentence structures.

The proposed **AI-Based Contextual Answer Evaluation System** is developed to assist teachers in evaluating digital descriptive answers using Artificial Intelligence, Natural Language Processing and Machine Learning techniques. The system analyses the contextual and semantic meaning of student answers by comparing them with reference answers using Sentence-BERT and cosine similarity. KeyBERT is used to identify important concepts, while rubric-based evaluation provides criterion-wise scoring. Machine Learning models such as Random Forest Regressor and XGBoost Regressor are used to support score prediction.

The system follows a teacher-in-the-loop approach where AI generates an initial evaluation and feedback, while the teacher reviews, modifies and approves the final result. This approach helps reduce evaluation time, improve consistency and provide meaningful feedback while maintaining human control over academic decisions. The proposed system aims to provide an efficient, transparent and reliable solution for contextual evaluation of descriptive answers.

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**ROSHINI V: 230701270**

**SAKSHI P: 230701279**

**SANDHIYA P: 230701282**

**SHRADHA S: 230701311**

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## LIST OF ABBREVIATION

| ABBREVIATION | DESCRIPTION                       |
|--------------|-----------------------------------|
| AI           | Artificial Intelligence           |
| API          | Application Programming Interface |
| DBMS         | Database Management System        |
| DFD          | Data Flow Diagram                 |
| JWT          | JSON Web Token                    |
| ML           | Machine Learning                  |
| MAE          | Mean Absolute Error               |
| NLP          | Natural Language Processing       |
| NLTK         | Natural Language Toolkit          |
| $R^2$        | Coefficient of Determination      |
| RBAC         | Role-Based Access Control         |
| REST         | Representational State Transfer   |
| RMSE         | Root Mean Squared Error           |
| SBERT        | Sentence-BERT                     |
| SQL          | Structured Query Language         |
| UI           | User Interface                    |
| UML          | Unified Modeling Language         |

# CHAPTER 1

## INTRODUCTION

### 1.1 GENERAL

The evaluation of descriptive and subjective answers is an important part of the educational process as it helps in assessing a student's understanding, knowledge, analytical ability and conceptual learning. Unlike objective questions, descriptive answers allow students to express their ideas and understanding in their own words. Traditionally, these answers are evaluated manually by teachers, who read each response, compare it with the expected answer and assign marks based on their knowledge and evaluation criteria. However, this process requires considerable time and effort, especially when a large number of students and answer scripts are involved. It may also lead to variations in evaluation and delays in providing results and feedback.

With the advancement of Artificial Intelligence (AI), Natural Language Processing (NLP) and Machine Learning (ML), technology can be used to assist in the evaluation of textual answers. Traditional automated answer evaluation systems mainly depend on keyword matching and basic text similarity techniques. However, these methods may not accurately identify correct answers when students use different words, synonyms or alternative sentence structures to express the same concept. Therefore, an effective automated evaluation system should analyze the contextual and semantic meaning of a student's response rather than depending only on exact word matching.

The proposed **AI-Based Contextual Answer Evaluation System** is designed to assist teachers in evaluating digital descriptive answers using Artificial Intelligence and Natural Language Processing techniques. The system compares student answers with reference answers using semantic similarity, identifies important concepts and performs rubric-based evaluation. Sentence-BERT is used for analyzing semantic relationships between answers, while KeyBERT helps in extracting important concepts. Machine Learning techniques are also used to support score prediction based on different evaluation features.

The proposed system follows a **teacher-in-the-loop approach**, where the AI system generates an initial evaluation, score and feedback, while the teacher reviews, modifies and approves the final result. This approach helps reduce the workload of teachers, improve consistency in evaluation and provide meaningful feedback to students while maintaining human control over important academic decisions.

## 1.2 OBJECTIVE

The main goal of the AI-Based Contextual Answer Evaluation System is to design and develop an intelligent answer evaluation framework that leverages Artificial Intelligence, Natural Language Processing and Machine Learning to overcome the limitations of traditional descriptive answer assessment. The objectives are outlined as follows:

1. **Contextual Answer Evaluation:** To develop a semantic analysis system using Sentence-BERT that understands the contextual meaning of student answers and compares them effectively with reference answers.
2. **Concept-Based Analysis:** To identify important concepts from reference answers using KeyBERT and analyze the concept coverage present in student responses.
3. **Rubric-Based Scoring:** To implement teacher-defined rubrics that enable criterion-wise evaluation and provide a transparent breakdown of the assigned score.
4. **Machine Learning-Based Prediction:** To use regression models such as Random Forest Regressor and XGBoost Regressor to predict scores based on semantic similarity, concept coverage and other evaluation features.
5. **Teacher-in-the-Loop Evaluation:** To enable teachers to review, modify and approve AI-generated scores and feedback, ensuring that final academic decisions remain under human supervision.
6. **Feedback and Performance Analysis:** To generate meaningful feedback for students and provide evaluation insights that help identify strengths and areas for improvement.

Achieving these objectives will provide a more efficient, consistent and transparent approach to descriptive answer evaluation while reducing the workload of teachers and maintaining the importance of human academic judgment.

## 1.3 EXISTING SYSTEM

The existing system for evaluating descriptive and subjective answers primarily relies on manual assessment by teachers. Teachers evaluate each student response by comparing it with their expected understanding of the subject and assigning marks accordingly. Although manual evaluation provides flexibility and human judgment, it becomes time-consuming and challenging when a large number of answer scripts need to be evaluated.

Some automated answer evaluation systems use keyword matching, text similarity and basic Natural Language Processing techniques. These approaches compare student answers with reference answers based on common words or phrases. However, they often fail to understand the contextual meaning of answers. Students may use different words, synonyms or sentence structures to express the same concept, which can result in inaccurate evaluation.

The existing systems also provide limited feedback and may lack transparency in the scoring process. Fully automated systems may not provide sufficient teacher control over the final evaluation.

## **1.4 PROPOSED SYSTEM**

The proposed AI-Based Contextual Answer Evaluation System is designed to overcome the limitations of traditional and keyword-based answer evaluation methods by using Artificial Intelligence, Natural Language Processing and Machine Learning techniques. The system evaluates digital descriptive answers based on their contextual and semantic meaning rather than relying only on exact keyword matching.

The proposed system uses Sentence-BERT to analyze the semantic similarity between student answers and reference answers. KeyBERT is used to extract important concepts, while teacher-defined rubrics enable criterion-wise evaluation. Machine Learning models such as Random Forest Regressor and XGBoost Regressor are used to support score prediction based on different evaluation features.

A major feature of the proposed system is the teacher-in-the-loop approach. The AI system generates an initial score, evaluation and feedback, while the teacher reviews, modifies and approves the final result. This ensures that Artificial Intelligence supports teachers rather than replacing human academic judgment.

### **Advantages of Proposed System**

- Provides contextual and semantic answer evaluation.
- Recognizes different expressions of the same concept.
- Supports concept extraction and rubric-based scoring.
- Uses Machine Learning for score prediction.
- Generates meaningful feedback for students.
- Reduces the time and effort required for evaluation.
- Improves consistency and transparency.
- Maintains teacher control over final academic decisions.

## CHAPTER 2

### LITERATURE SURVEY

The advancement of Artificial Intelligence, Natural Language Processing, and Machine Learning has significantly contributed to the development of intelligent educational assessment systems. Traditional descriptive answer evaluation methods mainly depend on manual assessment, which requires considerable time and effort and may lead to inconsistencies in scoring. In this context, research studies focusing on semantic similarity, transformer-based language models, automated short-answer grading, rubric-based evaluation, and human-in-the-loop assessment become highly relevant. This chapter reviews significant research works in these areas, highlighting their methodologies, strengths, limitations, and contributions to the development of the proposed AI-Based Contextual Answer Evaluation System.

#### 2.1 LITERATURE REVIEW

The paper **“LLM-based Automated Grading with Human-in-the-Loop”** by Li et al. (2025) presents an automated grading framework that combines the capabilities of Large Language Models with human expert involvement. The proposed GradeHITL framework focuses on improving the reliability and transparency of automated grading by allowing human evaluators to participate in refining grading rubrics and reviewing AI-generated assessments. Rather than completely replacing teachers, the framework uses human feedback to improve the evaluation process and address situations where the AI model may produce uncertain or inaccurate results. The study highlights that automated grading systems can benefit significantly from structured human supervision, particularly in subjective and descriptive assessments where contextual understanding is important. The framework also demonstrates the importance of transparent rubric-based evaluation instead of assigning a single unexplained score. However, the involvement of human experts may increase the time required for the overall evaluation process, especially when a large number of responses need detailed review. For the proposed AI-Based Contextual Answer Evaluation System, this work is highly relevant because it supports the implementation of a teacher-in-the-loop approach where AI provides an initial evaluation while teachers retain control over the final academic decision.

The paper **“Subjective Answers Evaluation Using Machine Learning and Natural Language Processing”** by Bashir et al. (2021) proposes an automated system for evaluating subjective answers using a combination of Natural Language Processing and Machine Learning techniques. The system uses approaches such as TF-IDF, Word2Vec, Word Mover's Distance, and cosine similarity to compare student

responses with reference answers. These techniques help the system move beyond simple keyword matching by analyzing the semantic relationship between words and sentences. The study evaluates different similarity methods and demonstrates that semantic approaches can provide better results when students use alternative words to express similar concepts. The proposed methodology also uses Machine Learning for predicting or classifying answer scores based on the extracted textual features. The study demonstrates the usefulness of semantic similarity in subjective answer evaluation; however, traditional embedding techniques may still have limitations in understanding complex contextual relationships within longer answers. For the proposed system, this research provides important insights into combining semantic similarity measures with Machine Learning-based score prediction and highlights the need for more contextual language representations.

The paper **“Automatic Short-Answer Grading via BERT-Based Deep Neural Networks”** by **Zhu et al. (2022)** investigates the use of transformer-based language models for automated short-answer grading. The proposed approach uses BERT to generate contextual representations of student and reference answers and combines deep learning architectures to capture semantic relationships more effectively. Unlike traditional text similarity methods, BERT considers the context in which words occur, enabling the model to distinguish between words with different meanings and understand relationships within a sentence. The study demonstrates that contextual embeddings can improve the performance of automated grading systems by recognizing answers that convey similar meanings using different vocabulary or sentence structures. The deep learning architecture further improves feature extraction for score prediction. Although transformer-based models provide strong contextual understanding, they require more computational resources compared to traditional NLP techniques and may require appropriate training data for effective performance. This research is particularly relevant to the proposed system because it supports the use of Sentence-BERT for generating semantic embeddings and comparing student answers with reference answers based on contextual meaning.

The paper **“GradeAid: A Framework for Automatic Short Answers Grading in Educational Contexts—Design, Implementation and Evaluation”** by **del Gobbo et al. (2023)** presents an automated grading framework that combines multiple linguistic and Machine Learning features for evaluating short answers. The proposed system incorporates lexical information, semantic representations, and regression-based models to predict student scores. The study demonstrates that relying on a single evaluation method may not be sufficient for accurately assessing descriptive responses and that combining multiple features can improve grading performance. The framework also focuses on providing a practical solution that can be applied in educational environments to reduce the repetitive workload of teachers. One of the major strengths of the study is its combination of traditional text features with semantic information generated through language models. However, the effectiveness of the system depends on the quality of the training

data and the suitability of the selected features for different subjects and question types. For the proposed AI-Based Contextual Answer Evaluation System, this work provides valuable guidance for combining semantic similarity, concept coverage, rubric scores, and other textual features for Machine Learning-based score prediction.

The paper **“Enhancing LLM-Based Short Answer Grading with Retrieval-Augmented Generation”** by **Chu et al. (2025)** explores the use of Retrieval-Augmented Generation to improve the performance of Large Language Models in automated short-answer grading. The proposed approach retrieves relevant domain-specific information and provides it to the language model during the evaluation process. This helps the model access additional contextual knowledge instead of relying only on the information available in its original training data. The study demonstrates that retrieval-based methods can improve answer evaluation when questions require specialized or subject-specific knowledge. The approach also highlights the importance of grounding AI-generated evaluations using relevant educational content. However, the performance of Retrieval-Augmented Generation depends on the quality and relevance of the retrieved information, and an unsuitable knowledge source may negatively affect the evaluation result. For the proposed system, this research provides useful insights for future enhancement through RAG-based feedback and domain-specific contextual evaluation, while the current system focuses on semantic similarity, concept extraction, and rubric-based assessment.

## **2.2 COMPARISON AND DISCUSSION**

The reviewed research works demonstrate the significant development of automated answer evaluation from traditional keyword-based approaches to advanced semantic and contextual analysis techniques. Earlier methods mainly relied on lexical matching, TF-IDF, word embeddings and syntactic analysis to compare student answers with reference answers. These approaches helped reduce the workload of educators but had limitations in understanding answers expressed using different vocabulary or sentence structures. Semantic similarity approaches improved this process by representing the meaning of answers rather than depending only on exact word matching.

Recent studies have increasingly adopted transformer-based models and Machine Learning techniques to improve contextual understanding and score prediction. Frameworks such as GradeAid demonstrate that combining lexical and semantic features with regression models can provide stronger automated grading performance than relying on a single feature. Similarly, research on deep learning approaches shows that semantic representations can work effectively when combined with carefully selected evaluation features. Another important observation from the literature is the growing importance of human involvement in automated assessment. While LLMs and advanced AI models provide improved language understanding,

completely automated grading may raise concerns regarding reliability, transparency and confidence in the assigned scores. Human-in-the-loop frameworks address this limitation by allowing uncertain or complex evaluations to be reviewed by human experts.

Based on the comparison of the reviewed studies, it is observed that an effective answer evaluation system should combine multiple techniques rather than depending on a single approach. The proposed AI-Based Contextual Answer Evaluation System therefore integrates Sentence-BERT for semantic similarity, KeyBERT for concept extraction, rubric-based scoring and Machine Learning models for score prediction. The teacher-in-the-loop approach further ensures that AI-generated evaluations are reviewed before final results are published. This combination aims to provide a more efficient, transparent and context-aware solution for descriptive answer evaluation.

## 2.3 CONCLUSION

The literature survey provided an understanding of the existing research and technological approaches used in automated descriptive and short-answer evaluation. The reviewed studies demonstrate the evolution of answer evaluation from traditional keyword-based and lexical approaches to semantic similarity, transformer-based language models, Machine Learning and Large Language Models.

The analysis shows that semantic and contextual understanding plays an important role in accurately evaluating student answers. Techniques such as word embeddings, BERT-based models and semantic similarity help identify correct answers even when students use different words or sentence structures. The studies also highlight the benefits of combining multiple evaluation features, including semantic similarity, lexical information and Machine Learning-based score prediction.

However, the literature also identifies challenges related to transparency, reliability and complete dependence on automated grading. Human involvement remains important, particularly in subjective assessments where academic judgment is required. Therefore, the proposed **AI-Based Contextual Answer Evaluation System** adopts a teacher-in-the-loop approach along with semantic analysis, concept extraction, rubric-based scoring and Machine Learning techniques.

Based on the findings from the literature survey, the proposed system aims to overcome the limitations of existing approaches by providing an efficient and context-aware evaluation process while maintaining teacher control over final academic decisions. The reviewed research provides the foundation for designing the methodology and architecture of the proposed system.

## **CHAPTER 3**

### **SYSTEM DESIGN**

The AI-Based Contextual Answer Evaluation System is designed as an intelligent web-based system that assists teachers in evaluating descriptive answers. The system integrates Artificial Intelligence, Natural Language Processing, semantic similarity analysis, concept extraction, rubric-based scoring, and Machine Learning to provide an effective evaluation process.

The system supports different users, including Administrators, Teachers, and Students, through role-based access. Teachers can create subjects, assessments, questions, reference answers, and evaluation rubrics. Students can submit their descriptive answers through the system. The submitted answers are processed by the AI evaluation engine, which analyses their semantic and contextual relationship with the corresponding reference answer.

The evaluation process uses Sentence-BERT to generate semantic representations and cosine similarity to measure the similarity between student and reference answers. KeyBERT is used to identify important concepts, while teacher-defined rubrics provide structured criterion-wise scoring. The generated evaluation is then reviewed by the teacher before the final result is published.

#### **3.1 SYSTEM ARCHITECTURE**

The system follows a three-layer architecture consisting of the Frontend Layer, Backend Layer, and Database and AI Engine Layer.

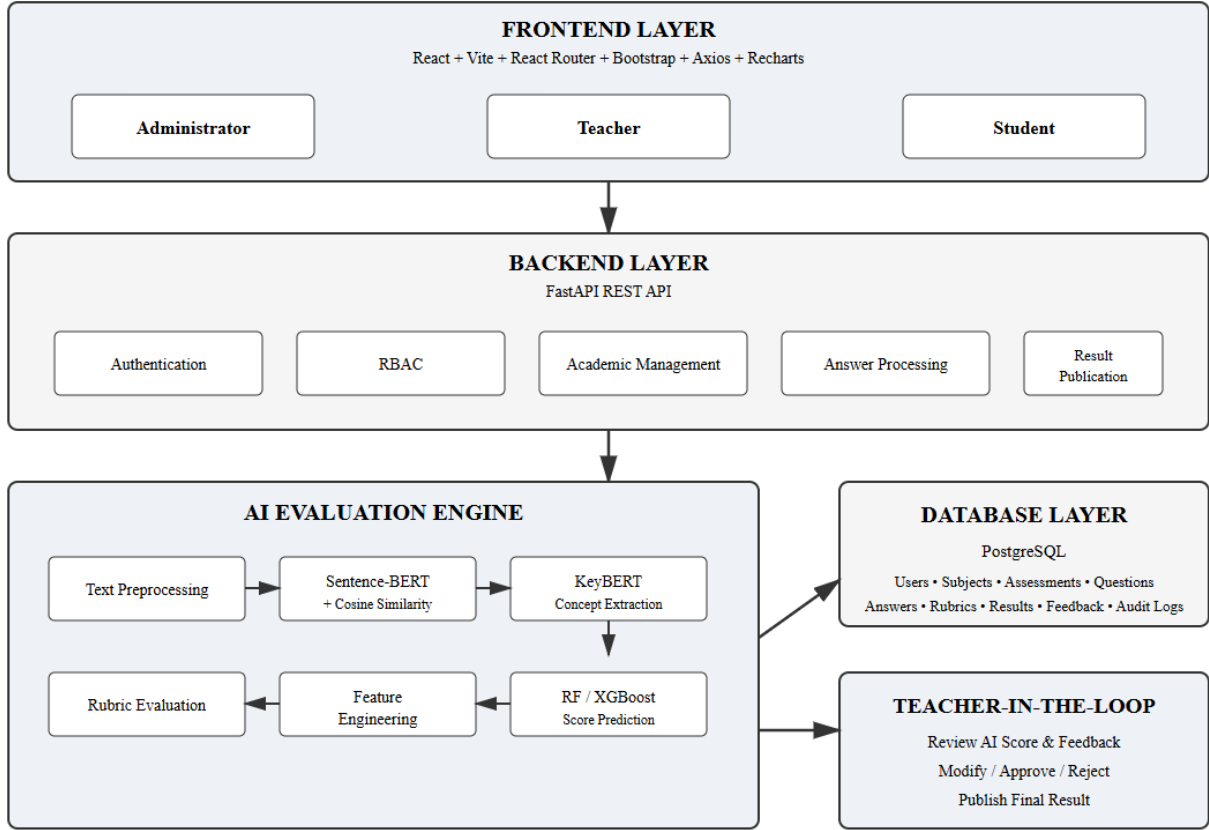
The Frontend Layer is developed using React and provides interfaces for administrators, teachers, and students. It allows users to access system functionalities based on their roles and communicates with the backend through REST APIs.

The Backend Layer is developed using FastAPI and manages authentication, user requests, academic data, answer submissions, evaluation processes, and result publication. JWT authentication is used to provide secure access to the system.

The AI Engine performs answer evaluation using Natural Language Processing and Machine Learning techniques. It processes student answers, calculates semantic similarity, extracts important concepts, performs rubric-based scoring, and predicts scores.

The Database Layer uses PostgreSQL for storing user information, subjects, assessments, questions, student answers, evaluation results, feedback, and audit records.

## AI-Based Contextual Answer Evaluation System



*Fig 3.1 Architecture Diagram*

### 3.2. SYSTEM REQUIREMENTS

The system requirements define the necessary hardware and software resources required for the development and execution of the proposed system. The system is designed to provide efficient answer evaluation while supporting multiple users and maintaining secure access to academic information.

The proposed system requires a computer system with sufficient processing capability to execute the web application, database operations, and AI-based evaluation processes. Internet connectivity is required for accessing the system when deployed as a web application.

To ensure smooth development, deployment, and operation of the AI-Based Contextual Answer Evaluation System, the following hardware and software specifications are recommended:

#### 3.2.1 Hardware Requirements

##### 1. Server/Processing Unit

- a. Multi-core processor (Intel i5/i7 or equivalent) for handling backend operations, Natural Language Processing, and Machine Learning-based answer evaluation.

- b. Minimum 8 GB RAM (16 GB recommended) for efficient execution of Sentence-BERT, KeyBERT, and Machine Learning models.
- c. SSD storage (256 GB or more) for storing application files, student answers, evaluation records, trained models, and databases.

## **2. User Devices**

- a. Desktop or Laptop for administrators and teachers to manage assessments, questions, rubrics, evaluations, and system activities.
- b. Desktop, Laptop, Tablet, or Smartphone for students to access assessments, submit descriptive answers, and view evaluation results and feedback.

## **3. Networking**

- a. Stable internet connectivity for communication between the frontend, backend, database, and AI evaluation services.
- b. Reliable network connection to support multiple users accessing the system and submitting answers simultaneously.

## **4. Storage and Database Infrastructure**

- a. Database server or cloud-based infrastructure for storing user information, subjects, assessments, questions, student answers, evaluation results, and audit records.
- b. Backup storage facilities to maintain data availability and support recovery of important academic information.

The hardware setup ensures that the AI-Based Contextual Answer Evaluation System can efficiently process descriptive answers while maintaining reliable system performance. By providing sufficient processing capability, memory, storage, and network connectivity, the system can support semantic analysis and Machine Learning-based evaluation without significant delays. The proposed hardware configuration also provides flexibility for future expansion, such as supporting more users, additional assessments, larger datasets, and advanced AI models.

### **3.2.2 Software Requirements**

#### **1. Operating System**

- a. Windows 10/11 or Linux (Ubuntu 20.04 or later) for the development and deployment environments.
- b. The operating system should support the execution of Python, React, PostgreSQL, and other required development tools.

#### **2. Development Tools and Languages**

- a. Python 3.12 for implementing the backend, Natural Language Processing, and Machine Learning modules.
- b. JavaScript and React for developing the user interface and web application.
- c. FastAPI for developing REST APIs and managing communication between the frontend, AI engine, and database.

#### **3. Libraries and Frameworks**

- a. Natural Language Processing: NLTK and Sentence-Transformers for text preprocessing and semantic analysis.
- b. Semantic Analysis: Sentence-BERT and cosine similarity for comparing student answers with reference answers.

- c. Concept Extraction: KeyBERT for identifying important concepts and analyzing concept coverage.
- d. Machine Learning: Scikit-learn and XGBoost for score prediction and evaluation.
- e. Data Processing: Pandas and NumPy for data manipulation and numerical operations.

#### **4. Database Management System**

- a. PostgreSQL for storing user information, subjects, assessments, questions, student answers, and evaluation results.
- b. SQLite for local development and testing.
- c. Support for database migration and management using SQLAlchemy and Alembic.

#### **5. Additional Tools**

- a. Version Control: Git for source code management and project version control.
- b. Containerization: Docker and Docker Compose for deployment and managing application services.
- c. Testing: Pytest and FastAPI TestClient for unit and integration testing.

The recommended software stack ensures that the AI-Based Contextual Answer Evaluation System operates efficiently and supports contextual answer analysis, Machine Learning-based score prediction, and secure web application functionality. By using modern frameworks, databases, and specialized NLP libraries, the system can process descriptive answers and provide meaningful evaluation results. Furthermore, this software configuration allows for easy integration of new features, updates to AI models, and future enhancements, ensuring that the system remains scalable and adaptable.

### **3.3. SYSTEM DESIGN**

The AI-Based Contextual Answer Evaluation System design model outlines how student answers transition from raw textual input to final evaluation results through structured processing stages. The system begins with answer submission and data management, where student responses, reference answers, and teacher-defined rubrics are collected and stored. The submitted answer then undergoes preprocessing to prepare the textual data for further analysis.

After preprocessing, the student answer is processed through the Semantic Analysis Module and Concept Extraction Module. The Semantic Analysis Module uses Sentence-BERT to generate contextual embeddings and calculates the relationship between the student answer and reference answer using cosine similarity. The Concept Extraction Module uses KeyBERT to identify important concepts and analyze their coverage within the student response.

The outputs from these modules are combined with rubric-based scores and other evaluation features. These features are processed by the Machine Learning Module, which uses Random Forest Regressor and XGBoost Regressor for score prediction. The generated score and feedback are then transferred to the Teacher Review Module, where the teacher can review, modify, approve, or reject the AI-generated

evaluation. This design ensures efficient evaluation, transparency, human supervision, and reliable result generation.

### **3.3.1 Activity Diagram**

The activity diagram represents the sequential workflow of the AI-Based Contextual Answer Evaluation System. It begins with the teacher creating an assessment and defining the questions, reference answers, and evaluation rubrics. Students then access the assessment and submit their descriptive answers through the system.

The submitted answer undergoes text preprocessing, followed by semantic analysis using Sentence-BERT and concept extraction using KeyBERT. The semantic similarity, concept coverage, rubric scores, and other evaluation features are then processed by the Machine Learning models for score prediction. The system generates an initial score and feedback based on the evaluation results.

The generated evaluation is then sent to the Teacher Review Module. The teacher examines the AI-generated score and feedback and decides whether to approve or modify the evaluation. Once the result is approved, it is published and made available to the student. The sequential workflow ensures that Artificial Intelligence supports the evaluation process while the final academic decision remains under teacher supervision.

## ACTIVITY DIAGRAM

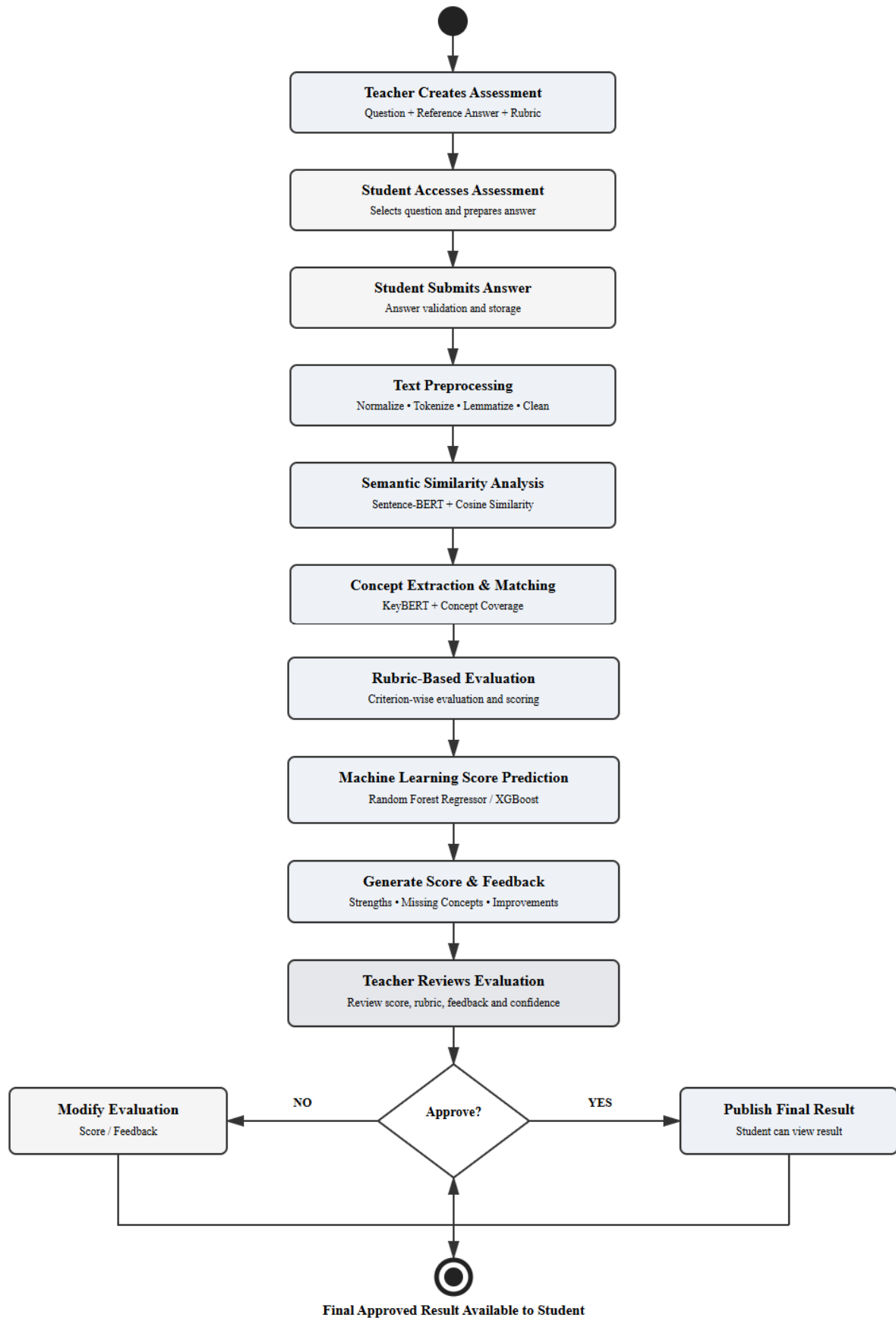


Fig 3.2 Activity Diagram

### 3.3.2 Data Flow Diagram

The Data Flow Diagram (DFD) represents the flow of information within the AI-Based Contextual Answer Evaluation System. It illustrates how data moves between the users, system modules, AI evaluation engine, and database during the answer evaluation process.

The system receives input from three main users: Administrator, Teacher, and Student. The administrator manages user accounts and system-related information. The teacher creates subjects, assessments, questions, reference answers, and evaluation rubrics. The student accesses the assessment and submits descriptive answers through the system.

The submitted student answer is processed by the AI Evaluation Engine. The answer undergoes text preprocessing, semantic similarity analysis, and concept extraction. Sentence-BERT is used to analyze the contextual similarity between the student answer and reference answer, while KeyBERT identifies important concepts. The generated evaluation features are combined with rubric scores and processed using Machine Learning models for score prediction.

The generated score and feedback are sent to the Teacher Review Module. The teacher can review, modify, and approve the AI-generated evaluation. Once approved, the final result is stored in the Database and published to the student.

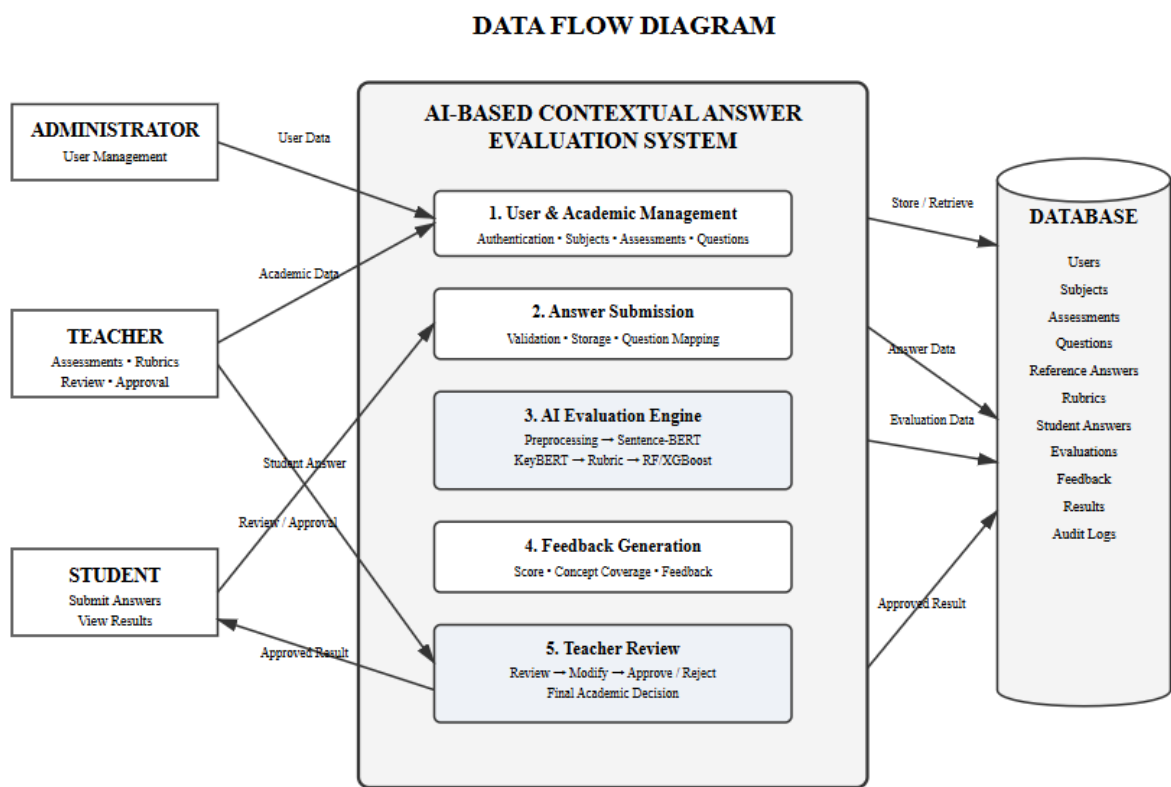


Fig 3.3 Data Flow Diagram

## **CHAPTER 4**

### **PROJECT DESCRIPTION**

#### **4.1 METHODOLOGY**

The proposed **AI-Based Contextual Answer Evaluation System** follows a structured methodology to evaluate descriptive answers using **Natural Language Processing, Semantic Analysis, Concept Extraction, Rubric-Based Evaluation, and Machine Learning**. The methodology is designed to analyze the contextual meaning of student answers and provide an initial evaluation while maintaining teacher involvement in the final assessment process.

The complete methodology begins with the creation of an assessment by the teacher. The teacher defines the question, provides a reference answer, and creates an appropriate evaluation rubric. Students access the assessment through the system and submit their descriptive answers. The submitted answer is then forwarded to the AI evaluation engine for further processing.

The AI engine performs text preprocessing to prepare the student response for analysis. The processed answer is compared with the reference answer using semantic similarity techniques. Important concepts are extracted and analyzed to determine whether the expected concepts are present in the student's response. The evaluation results are combined with rubric-based scores and additional features for Machine Learning-based score prediction.

Finally, the system generates an initial score and feedback. The generated result is sent to the teacher for review and approval. Only after teacher approval is the final result published to the student. This methodology ensures an efficient and structured evaluation process while maintaining human supervision and transparency.

#### **4.2 MODULES**

The AI-Based Contextual Answer Evaluation System consists of several interconnected modules designed to manage users, assessments, answer submissions, AI-based evaluation, teacher review, and result publication. Each module performs a specific function and communicates with other modules to ensure a structured and efficient evaluation process.

## **4.2.1 AUTHENTICATION AND USER MANAGEMENT MODULE**

The Authentication and User Management Module manages user registration, login, and secure access to the system. The system supports different users, including Administrator, Teacher, and Student, with separate functionalities based on their roles.

The authentication process uses JWT-based authentication to provide secure access to system resources. User credentials are validated during login, and authorized users are provided access according to their assigned roles.

### **Submodules**

1. User Registration
2. User Login
3. JWT Authentication
4. Role-Based Access Control
5. Profile Management

This module ensures secure access and prevents unauthorized users from accessing restricted academic and evaluation information.

## **4.2.2 ACADEMIC STRUCTURE MANAGEMENT MODULE**

The Academic Structure Management Module manages the basic academic information required for conducting assessments. Teachers can create and organize subjects, assessments, and questions within the system.

The module provides a structured relationship between different academic entities. Each assessment can contain multiple questions, and each question can have a corresponding reference answer and evaluation rubric.

### **Submodules**

1. Subject Management
2. Assessment Management
3. Question Management
4. Reference Answer Management

This module provides an organized structure for managing academic assessments and evaluation-related information.

### **4.2.3 RUBRIC MANAGEMENT MODULE**

The Rubric Management Module allows teachers to define evaluation criteria for descriptive answers. A rubric provides a structured method for assigning marks based on different aspects of an expected answer. Teachers can create multiple criteria and allocate appropriate marks for each criterion. The AI evaluation system uses the rubric information to perform criterion-wise analysis and generate a structured evaluation.

#### **Submodules**

1. Rubric Creation
2. Criterion Definition
3. Mark Allocation
4. Rubric Modification

The use of rubrics improves transparency, consistency, and explainability in the answer evaluation process.

### **4.2.4 STUDENT ANSWER SUBMISSION MODULE**

The Student Answer Submission Module enables students to access available assessments and submit their descriptive answers through the application. The system displays the required questions and provides an interface for entering answers.

Before processing, the submitted answer is validated and stored in the database along with the corresponding student, assessment, and question information. The answer is then forwarded to the AI Evaluation Module.

#### **Submodules**

1. Assessment Access
2. Question Display
3. Answer Submission
4. Submission Validation
5. Answer Storage

This module acts as the primary input stage for the AI-based evaluation process.

### **4.2.5 AI-BASED ANSWER EVALUATION MODULE**

The AI-Based Answer Evaluation Module is the core module of the proposed system. It processes the submitted student answer and compares it with the reference answer using various Artificial Intelligence and Natural Language Processing techniques.

The module performs preprocessing, semantic analysis, concept extraction, rubric evaluation, and score prediction. The results generated by these processes are combined to provide an initial evaluation.

### **Submodules**

1. Text Preprocessing
2. Semantic Analysis
3. Concept Extraction
4. Rubric Evaluation
5. Feature Generation
6. Score Prediction

This module enables the system to analyze answers based on their contextual meaning rather than relying only on exact keyword matching.

## **4.2.6 SEMANTIC SIMILARITY MODULE**

The Semantic Similarity Module evaluates the contextual relationship between the student answer and the reference answer. The module uses Sentence-BERT to convert textual answers into numerical vector representations called embeddings.

The embeddings of the student and reference answers are compared using Cosine Similarity. The resulting similarity score represents the semantic relationship between the two answers.

### **Submodules**

1. Student Answer Embedding
2. Reference Answer Embedding
3. Sentence-BERT Processing
4. Cosine Similarity Calculation

This module helps identify answers with similar meanings even when different words or sentence structures are used.

## **4.2.7 CONCEPT EXTRACTION AND MATCHING MODULE**

The Concept Extraction and Matching Module identifies important concepts from the reference answer and analyses their presence in the student response. The system uses KeyBERT to extract important keywords and concepts.

The extracted concepts are compared with the student answer to determine the level of concept coverage. This helps evaluate whether the student has included the important ideas required for answering the question.

### **Submodules**

1. Keyword Extraction
2. Concept Identification
3. Concept Matching
4. Concept Coverage Calculation

The concept coverage result provides an additional evaluation measure along with semantic similarity.

#### **4.2.8 MACHINE LEARNING-BASED SCORE PREDICTION MODULE**

The Machine Learning-Based Score Prediction Module predicts the score of a student answer using the features generated during the evaluation process. Features such as semantic similarity, concept coverage, rubric scores, word count, and sentence count are used as inputs.

The system uses Random Forest Regressor and XGBoost Regressor for numerical score prediction. The performance of the models is evaluated using appropriate evaluation metrics, and the better-performing model can be selected.

##### **Submodules**

1. Feature Collection
2. Model Training
3. Score Prediction
4. Model Evaluation
5. Model Selection

This module provides a data-driven approach to support answer score prediction.

#### **4.2.9 FEEDBACK GENERATION MODULE**

The Feedback Generation Module generates meaningful feedback based on the evaluation results. The system analyses semantic similarity, concept coverage, rubric scores, and predicted performance to identify the strengths and weaknesses of the student's answer.

The feedback can indicate missing concepts, partially covered ideas, and areas requiring improvement. This enables students to understand their performance in addition to viewing the final score.

##### **Submodules**

1. Strength Identification
2. Missing Concept Detection
3. Improvement Suggestions
4. Feedback Generation

This module supports a more effective learning experience by providing personalized evaluation information.

#### **4.2.10 TEACHER REVIEW AND APPROVAL MODULE**

The Teacher Review and Approval Module implements the teacher-in-the-loop approach of the proposed system. AI-generated scores and feedback are not directly published to students.

The teacher reviews the evaluation results and can modify the score, edit the feedback, approve the result, or reject the evaluation when necessary. This ensures that the final academic decision remains under human supervision.

##### **Submodules**

1. Evaluation Review
2. Score Modification
3. Feedback Editing
4. Result Approval
5. Evaluation Rejection

This module combines the efficiency of AI with the experience and academic judgment of teachers

#### **4.2.11 RESULT PUBLICATION MODULE**

The Result Publication Module manages the final evaluation results after teacher approval. The approved score, criterion-wise evaluation, and feedback are stored in the system and made available to the respective student.

The system ensures that only teacher-approved results are published. Students can access their results through their individual accounts.

##### **Submodules**

1. Final Score Storage
2. Approval Status Management
3. Result Publication
4. Student Result View

This module ensures a controlled and reliable process for releasing evaluation results.

# **CHAPTER 5**

## **IMPLEMENTATION AND RESULT DISCUSSION**

### **5.1 IMPLEMENTATION PLAN**

The implementation plan describes the systematic process followed to develop and deploy the AI-Based Contextual Answer Evaluation System. The implementation is divided into different phases to ensure that each component of the system is developed, integrated, and tested properly. The system combines web development, database management, Natural Language Processing, and Machine Learning techniques to provide an efficient answer evaluation process.

The implementation begins with requirement analysis and system setup, followed by the development of the frontend, backend, database, and AI evaluation modules. After integrating all components, the system is tested using different student answers and evaluation scenarios. The following sections describe the implementation plan in detail.

#### **5.1.1 REQUIREMENT ANALYSIS AND PLANNING**

The first phase of implementation involves identifying the functional and technical requirements of the proposed system. The requirements are analysed based on the need to evaluate descriptive answers efficiently while maintaining teacher involvement in the final evaluation process.

The major functional requirements identified include user authentication, role-based access, subject and assessment management, question creation, reference answer management, rubric creation, student answer submission, AI-based evaluation, teacher review, and result publication.

The technical requirements include the selection of suitable technologies for frontend development, backend services, database management, Natural Language Processing, and Machine Learning. React is selected for the frontend, FastAPI for the backend, and PostgreSQL for database management. The AI evaluation process is implemented using Python, Sentence-BERT, KeyBERT, and Machine Learning models.

Proper requirement analysis ensures that the implementation process follows a structured approach and that all modules meet the objectives of the proposed system.

### **5.1.2 DEVELOPMENT ENVIRONMENT SETUP**

The development environment is configured before implementing the individual system modules.

The required software, frameworks, libraries, and development tools are installed and configured. The frontend development environment is created using React and Vite. Required libraries such as React Router, Bootstrap, Axios, and Recharts are integrated into the application.

The backend environment is configured using Python and FastAPI. Additional libraries such as SQLAlchemy, Pydantic, and authentication libraries are installed for API development, database communication, and data validation.

The Artificial Intelligence environment includes NLTK, Sentence-Transformers, KeyBERT, scikit-learn, XGBoost, NumPy, and Pandas. These libraries support text preprocessing, semantic analysis, concept extraction, feature processing, and score prediction.

The database environment is configured using PostgreSQL, while SQLite can be used during local development and testing.

### **5.1.3 DATABASE IMPLEMENTATION**

The database implementation phase focuses on creating the data storage structure required for managing users, academic information, answers, and evaluation results. The proposed system uses a relational database to maintain structured relationships between different entities.

The database includes entities such as User, Subject, Assessment, Question, Reference Answer, Rubric, Rubric Criterion, Student Answer, Evaluation, Criterion Evaluation, Feedback, Teacher Review, Result, and Audit Log.

Relationships are established between the entities to ensure proper data management. For example, a subject can contain multiple assessments, an assessment can contain multiple questions, and each question can have a reference answer and multiple rubric criteria.

The database is integrated with the backend using SQLAlchemy ORM, which enables Python objects to interact with database tables. Alembic is used to manage database schema changes and migrations.

### **5.1.4 FRONTEND IMPLEMENTATION**

The frontend is implemented using React to provide an interactive and user-friendly interface for the system users. The interface is designed separately for Administrators, Teachers, and Students based on their system roles.

The frontend provides pages for user authentication, dashboard access, subject management, assessment creation, question management, answer submission, evaluation review, result viewing, and analytics.

React Router is used to manage navigation between different pages. Axios is used for communication between the frontend and backend APIs. Bootstrap and Bootstrap Icons are used to design responsive and consistent user interfaces.

The frontend communicates with the FastAPI backend through REST APIs. Data received from the backend is displayed to users according to their roles and permissions.

### 5.1.5 BACKEND IMPLEMENTATION

The backend is implemented using FastAPI, which provides RESTful APIs for communication between the frontend, database, and AI evaluation engine.

The backend handles user authentication, role-based access control, academic data management, assessment creation, answer submission, evaluation processing, teacher review, and result publication.

Pydantic is used to validate incoming and outgoing data. This ensures that user inputs and API requests follow the required data structure. SQLAlchemy manages database operations, while JWT tokens are used to maintain secure user sessions.

The backend also acts as a bridge between the database and the AI evaluation engine. It sends the required student answer, reference answer, and rubric information to the AI module and receives the generated evaluation results.

### 5.1.6 AUTHENTICATION AND ROLE-BASED ACCESS IMPLEMENTATION

The authentication module is implemented to provide secure access to the system. Users can log in using valid credentials, and the system verifies their information before granting access.

The system uses JSON Web Tokens (JWT) for authentication. After successful login, the backend generates a token that is used to verify authorised user requests.

Role-Based Access Control is implemented to provide different permissions to different users.

- **Administrator** manages users and system-level information.
- **Teacher** manages subjects, assessments, questions, rubrics, and evaluation reviews.
- **Student** accesses assessments, submits answers, and views approved results.

This implementation ensures that users can only access the features and information permitted for their respective roles.

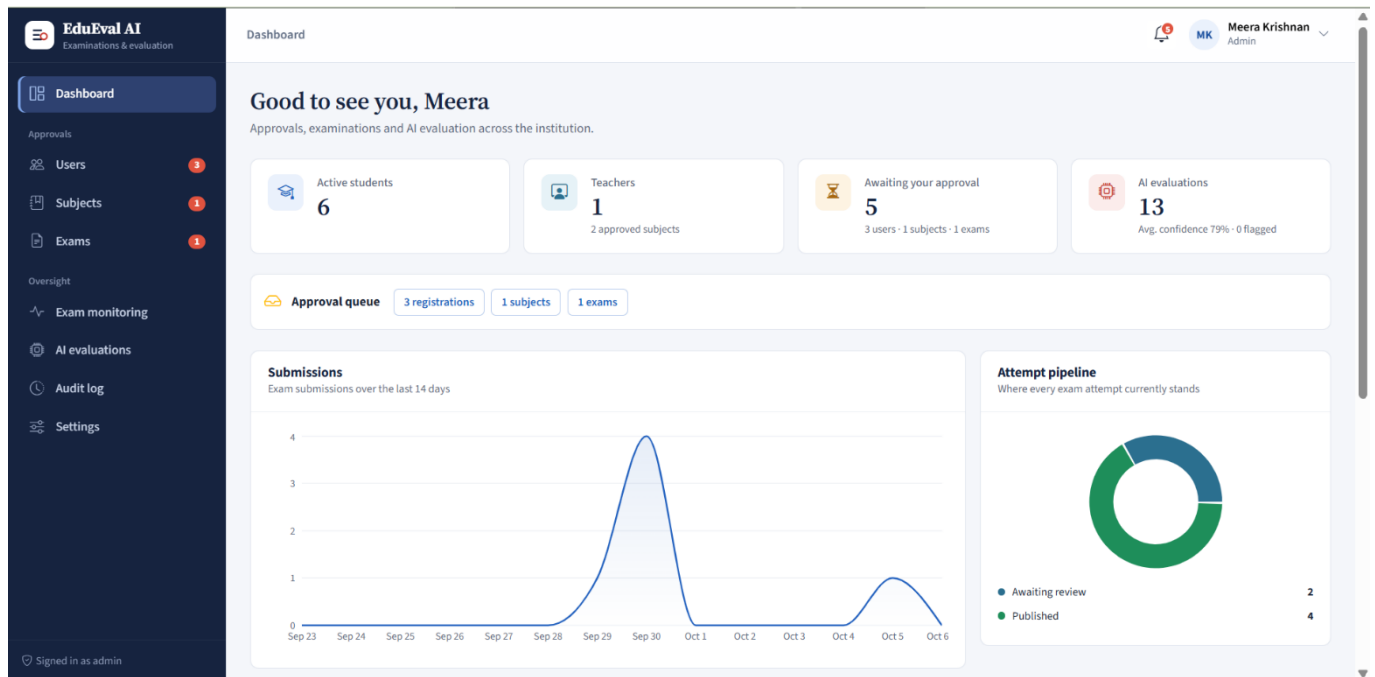


Fig 5.1 Admin Dashboard

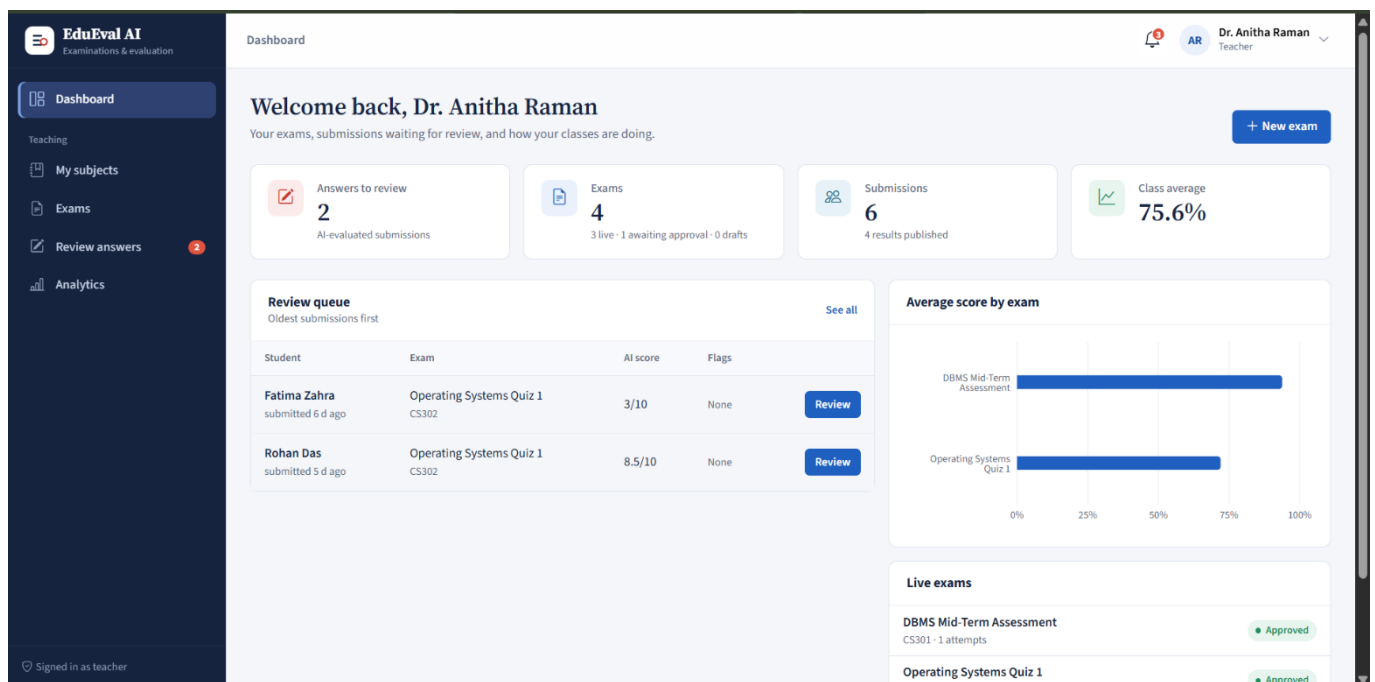
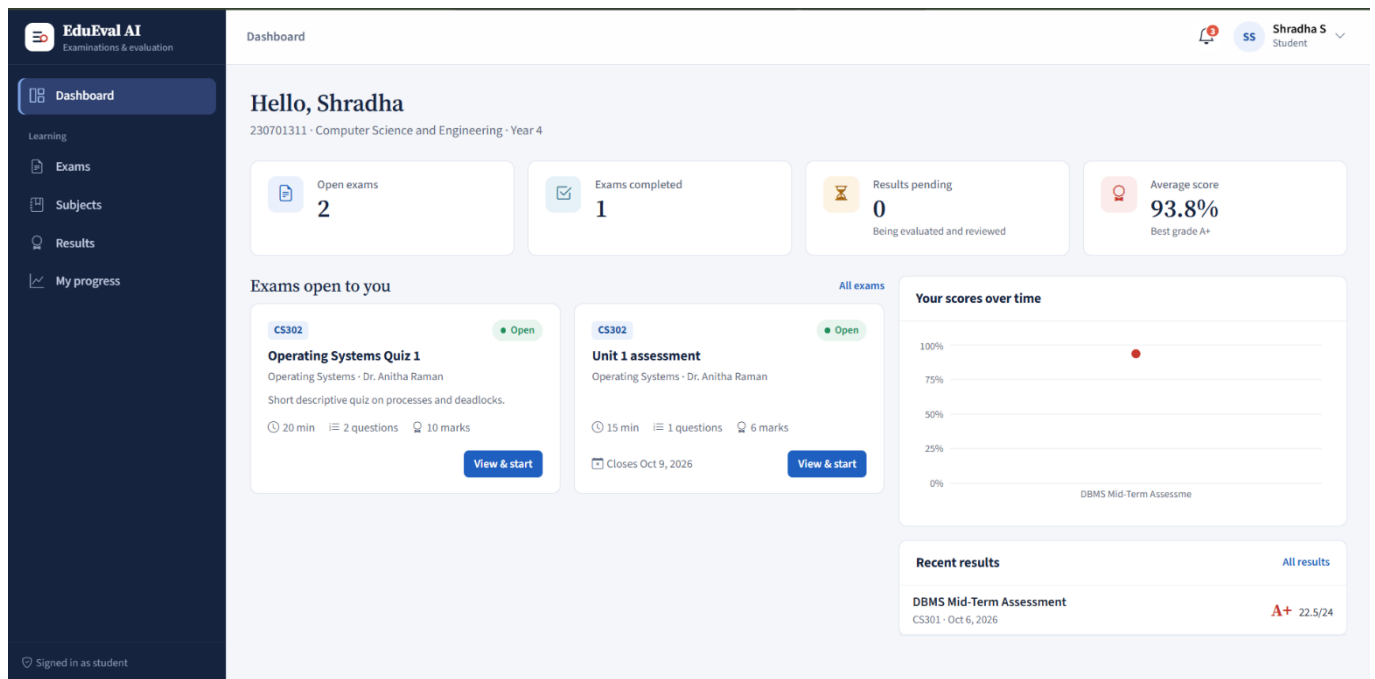


Fig 5.2 Teacher Dashboard



*Fig 5.3 Student Dashboard*

## 5.1.7 ASSESSMENT AND RUBRIC IMPLEMENTATION

The assessment module enables teachers to create and manage assessments. Teachers can add questions, specify maximum marks, and provide reference answers.

For each question, teachers can define an evaluation rubric containing different criteria and mark allocations. Each criterion represents an important aspect of the expected answer.

The rubric information is stored in the database and retrieved during the evaluation process. This enables the system to perform structured and criterion-wise evaluation instead of generating only a single score.

The assessment and rubric modules provide the foundation required for transparent and controlled answer evaluation.

**EduEval AI**  
Examinations & evaluation

Exams

Unit 1 a  
CS302 0 ques

Questions (0)

What is process scheduling in an Operating System? Explain any three process scheduling algorithms with suitable examples.

Marks: 6 Expected length (words): 150

**Reference answer**

Process scheduling is the mechanism used by an Operating System to select a process from the ready queue and allocate the CPU to it for execution. The main objective of process scheduling is to improve CPU utilization, reduce waiting time and response time, and ensure fair execution of processes. FCFS (First Come First Serve) executes processes in the order they arrive in the ready queue. It is simple but may result in a long waiting time for shorter processes. SJF (Shortest Job First) selects the process with the shortest CPU burst time and generally provides

Write it as a strong student would. Key concepts are extracted from it automatically.

**Must-have concepts (optional)**

ready queue, CPU, burst time, time quantum, waiting time

**Rubric criteria** 6 / 6 marks allocated Split evenly + Add criterion

| Definition                                                                                       | ready queue, CPU, scheduler             | 1.5 |
|--------------------------------------------------------------------------------------------------|-----------------------------------------|-----|
| Defines process scheduling as the OS deciding which process in the ready queue gets the CPU next |                                         |     |
| FCFS                                                                                             | first come first served, arrival, order | 1.5 |
| Explains First Come First Served with a correct example                                          |                                         |     |
| SJF                                                                                              | shortest job, burst time, waiting time  | 1.5 |
| Explains Shortest Job First using burst time with a correct example                              |                                         |     |
| Round Robin                                                                                      | round robin, time quantum, preempt      | 1.5 |
| Explains Round Robin using a time quantum and preemption with a correct example                  |                                         |     |

Cancel Save question

Delete Submit for approval

Signed in as teacher

*Fig 5.4 Questions with reference answer and rubric criteria*

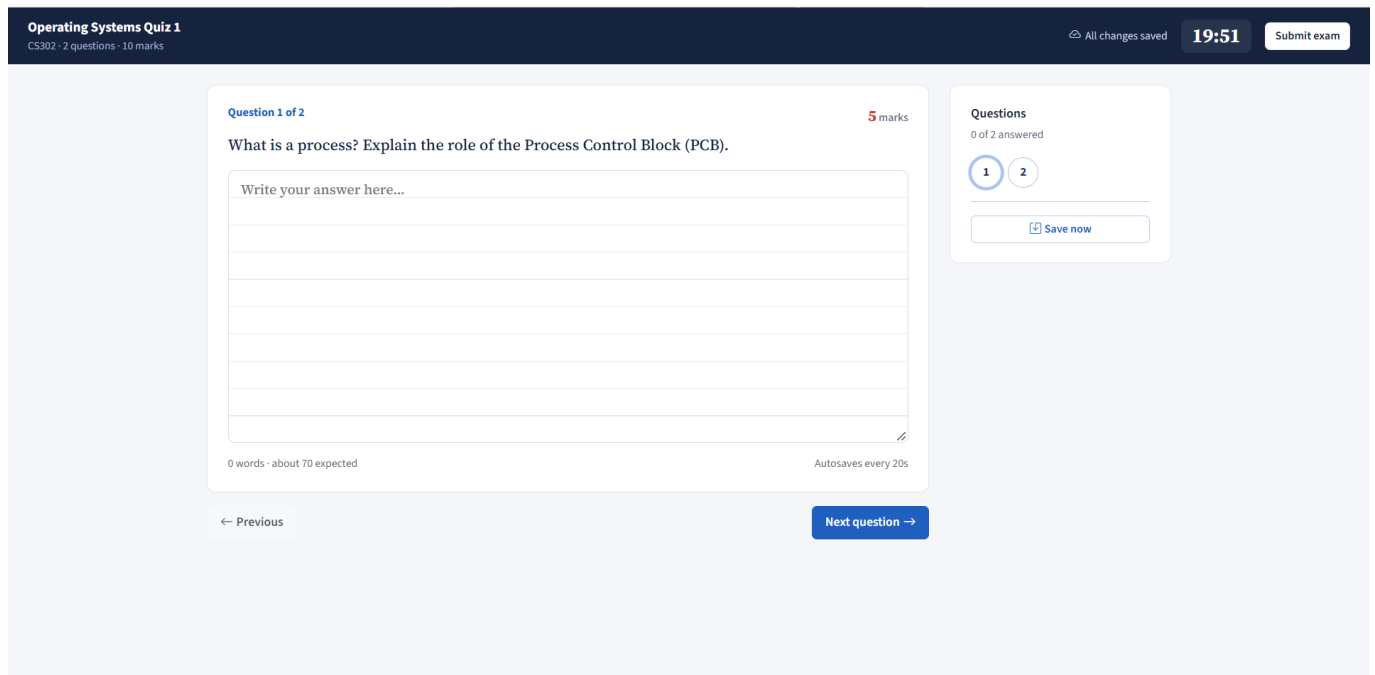
### 5.1.8 ANSWER SUBMISSION IMPLEMENTATION

The student answer submission module provides an interface through which students can access questions and submit descriptive responses.

When an answer is submitted, the system validates the input and associates it with the correct student, assessment, and question. The submitted response is stored securely in the database.

The backend then prepares the required information for AI evaluation. This includes the student answer, reference answer, question information, and rubric criteria.

The answer is then forwarded to the AI evaluation engine for further processing.



*Fig 5.5 Exam interface with timer and auto save*

## 5.1.9 TEXT PREPROCESSING IMPLEMENTATION

The submitted student answer and reference answer are processed before performing semantic analysis. Text preprocessing helps remove unnecessary variations and prepares the text for Natural Language Processing.

The preprocessing stage may include:

- Text normalization
- Lowercase conversion
- Tokenization
- Stop-word processing
- Lemmatization
- Whitespace removal

The preprocessing operations are implemented using Python and NLP libraries such as NLTK.

The processed text is then used for semantic similarity analysis and concept extraction. This stage improves the consistency and quality of the input provided to the AI models.

## 5.1.10 SEMANTIC ANALYSIS IMPLEMENTATION

The semantic analysis module is implemented using Sentence-BERT (SBERT) with the all-MiniLM-L6-v2 model.

The student answer and reference answer are converted into semantic vector representations called sentence embeddings. These embeddings represent the contextual meaning of the text.

The embeddings are compared using cosine similarity. The similarity score indicates how closely the student's answer matches the meaning of the reference answer.

A high similarity score indicates a strong semantic relationship, while a lower score may indicate that the student's answer does not sufficiently match the expected concept.

This implementation enables the system to recognize correct answers even when students use different words or sentence structures.

### **5.1.11 CONCEPT EXTRACTION IMPLEMENTATION**

The concept extraction module is implemented using KeyBERT. The system extracts important keywords and concepts from the reference answer.

The extracted concepts represent the major ideas that are expected to be present in a correct student response. The system analyses the student answer to determine the level of coverage of these concepts.

The implementation generates a concept coverage score, which represents the extent to which important concepts are addressed in the student answer.

Concept extraction provides additional information that complements semantic similarity analysis. An answer may have general semantic similarity but still miss important concepts. Therefore, concept analysis helps improve the completeness of evaluation.

### **5.1.12 RUBRIC-BASED EVALUATION IMPLEMENTATION**

The rubric-based evaluation module processes the criteria defined by the teacher for each question. Each rubric criterion contains information about:

- Criterion name
- Criterion description
- Maximum marks
- Expected concept or requirement

The student answer is evaluated against the defined criteria. The system generates criterion-wise scores based on the available evaluation information.

The criterion-wise results provide a detailed explanation of how the overall score is generated. This improves transparency and helps teachers review individual aspects of the evaluation.

### 5.1.13 FEATURE ENGINEERING IMPLEMENTATION

The outputs generated from different AI evaluation stages are combined to create a set of features for Machine Learning-based score prediction.

The major features include:

- Semantic similarity score
- Concept coverage score
- Rubric-based score
- Word count
- Sentence count

The feature values are processed using NumPy and Pandas.

The generated feature set is used as input for the regression models. Combining multiple features allows the system to analyze different aspects of answer quality instead of relying on only one evaluation method.

### 5.1.14 MACHINE LEARNING MODEL IMPLEMENTATION

The Machine Learning implementation uses regression algorithms to predict numerical scores based on the extracted evaluation features.

The proposed system uses:

#### **Random Forest Regressor**

Random Forest Regressor is used to analyze relationships between the evaluation features and expected scores. The model combines multiple decision trees to generate predictions.

#### **XGBoost Regressor**

XGBoost Regressor is used as an additional score prediction model. It applies gradient boosting techniques to improve prediction performance.

The models are evaluated using:

- **Mean Absolute Error (MAE)**
- **Root Mean Squared Error (RMSE)**
- **R<sup>2</sup> Score**

The model with better performance can be selected for score prediction. The Machine Learning implementation can also be improved over time using teacher-approved evaluation data.

### 5.1.15 FEEDBACK GENERATION IMPLEMENTATION

The feedback generation module uses the evaluation results to provide meaningful information about student performance.

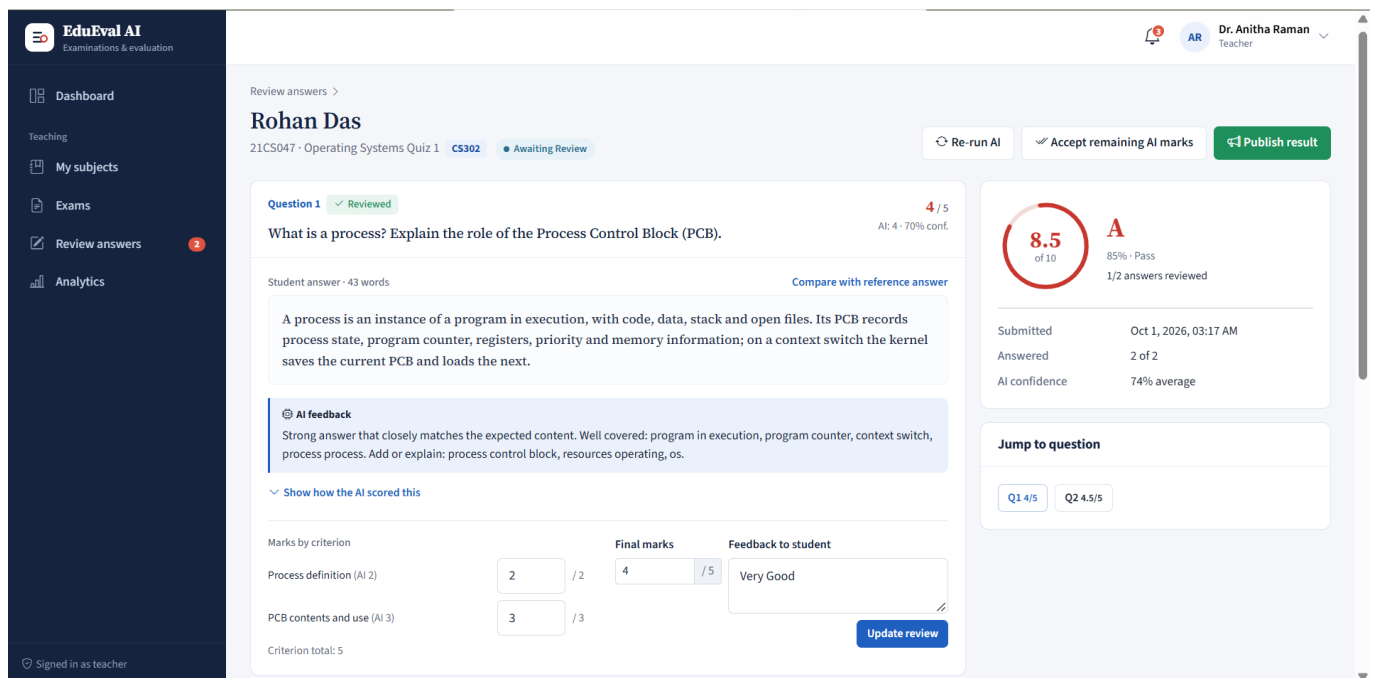
The system analyses:

- Semantic similarity
- Concept coverage
- Rubric performance
- Predicted score

Based on these results, feedback is generated to identify strengths, missing concepts, and areas requiring improvement.

For example, the system can indicate whether the student's answer has addressed the important concepts but requires additional explanation or whether major concepts are missing.

The generated feedback is initially provided to the teacher for review before it is published to the student.



The screenshot displays the 'Review answers' page for 'Rohan Das' in the '21CS047 - Operating Systems Quiz 1' (CS302). The interface is for a teacher, Dr. Anitha Raman. The question is 'What is a process? Explain the role of the Process Control Block (PCB)'. The student's answer is 43 words long. The AI feedback states: 'Strong answer that closely matches the expected content. Well covered: program in execution, program counter, context switch, process process. Add or explain: process control block, resources operating, os.' The AI score is 4/5 with 70% confidence. The teacher can override the marks by criterion: Process definition (AI 2) is 2/2, PCB contents and use (AI 3) is 3/3, and the final marks are 4/5. The feedback to the student is 'Very Good'. The overall criterion total is 5. The right sidebar shows a score of 8.5/10, an 'A' grade, 85% pass rate, and submission details for Oct 1, 2026, 03:17 AM.

*Fig 5.6 Teacher mark override and feedback generation*

## 5.1.16 TEACHER REVIEW IMPLEMENTATION

The teacher review module is implemented to ensure that the system follows a human-in-the-loop approach.

After AI evaluation, the teacher can view:

- AI-generated score
- Semantic similarity result
- Concept coverage
- Rubric-wise evaluation
- Generated feedback

- Confidence information

The teacher can then approve the evaluation or make necessary modifications to the score and feedback.

This implementation ensures that the final evaluation is not completely dependent on Artificial Intelligence. Teacher decisions remain the final authority before result publication.

| Student                      | Exam                              | Submitted | Reviewed | Score   | AI confidence | Status                    |
|------------------------------|-----------------------------------|-----------|----------|---------|---------------|---------------------------|
| Shradha S<br>230701311       | DBMS Mid-Term Assessment<br>CS301 | 9 h ago   | 3/3      | 22.5/24 | 85%           | Published<br>Open         |
| Rohan Das<br>21CS047         | Operating Systems Quiz 1<br>CS302 | 5 d ago   | 0/2      | 8.5/10  | 74%           | Awaiting Review<br>Review |
| Fatima Zahra<br>21CS031      | Operating Systems Quiz 1<br>CS302 | 6 d ago   | 0/2      | 3/10    | 77%           | Awaiting Review<br>Review |
| Karthik Natarajan<br>21CS022 | Operating Systems Quiz 1<br>CS302 | 6 d ago   | 2/2      | 8/10    | 80%           | Published<br>Open         |
| Priya Sharma<br>21CS014      | Operating Systems Quiz 1<br>CS302 | 6 d ago   | 2/2      | 7.5/10  | 71%           | Published<br>Open         |
| Arjun Kumar<br>21CS001       | Operating Systems Quiz 1<br>CS302 | 6 d ago   | 2/2      | 9/10    | 81%           | Published<br>Open         |

*Fig 5.7 Students submissions and review*

### 5.1.17 RESULT PUBLICATION IMPLEMENTATION

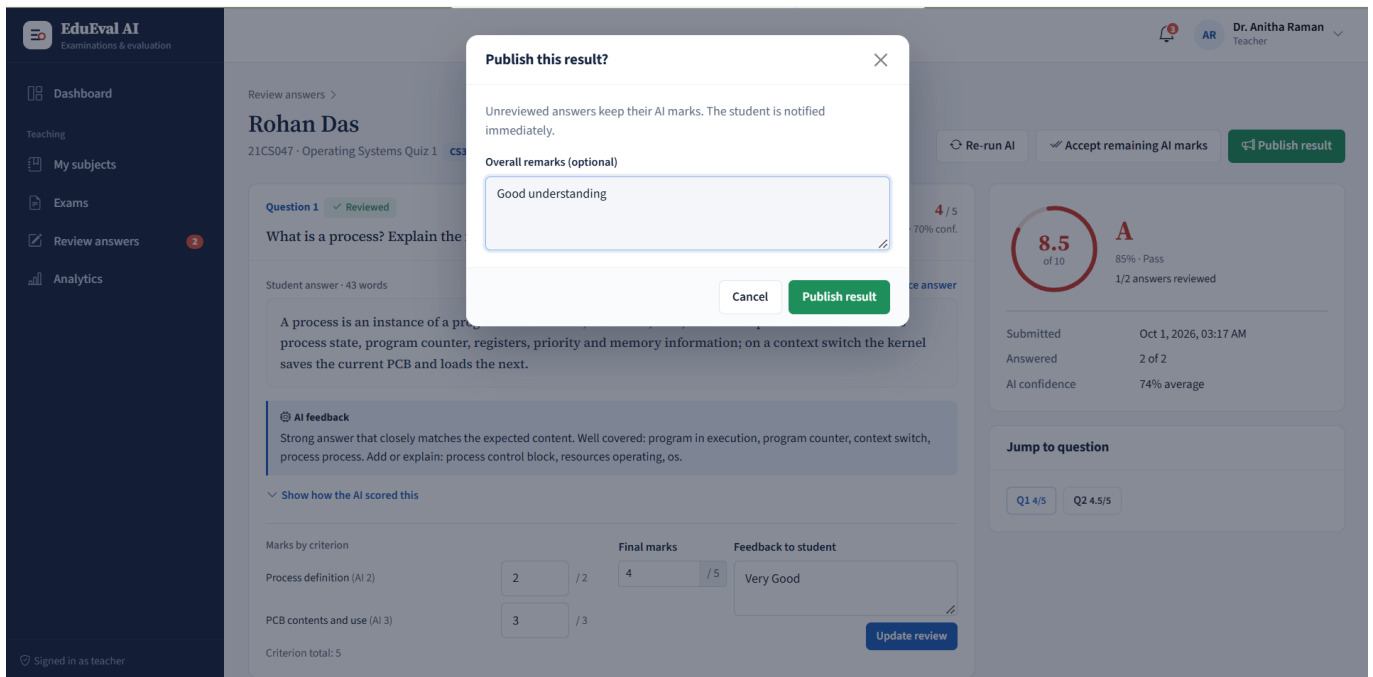
The result publication module manages the release of approved evaluation results.

Only after teacher approval, the final score and feedback are stored as the official result. Students can access the published result through their accounts.

The result may include:

- Final Score
- Maximum Marks
- Rubric-wise Evaluation
- Feedback
- Teacher Approval Status

This controlled process ensures that incomplete or unreviewed AI-generated evaluations are not directly published to students.



*Fig 5.8 Publishing results*

### 5.1.18 TESTING AND INTEGRATION

After implementing the individual modules, all components are integrated to ensure that the complete system works correctly.

The testing process includes:

- Unit Testing
- Integration Testing
- Authentication Testing
- Role-Based Access Testing
- AI Module Testing
- API Testing

Pytest and FastAPI TestClient are used for testing backend and API functionality.

The complete workflow is tested from student answer submission → AI evaluation → teacher review → result publication. Testing ensures that data is correctly transferred between the frontend, backend, AI engine, and database.

### 5.1.19 DEPLOYMENT IMPLEMENTATION

The system is prepared for deployment using Docker and Docker Compose. Containerization enables the frontend, backend, and database services to run in separate but connected environments. The main deployment components include:

- React Frontend Container

- FastAPI Backend Container
- PostgreSQL Database Container

Docker improves portability and ensures that the system can be deployed consistently across different environments.

The implementation plan provides a systematic approach for developing the AI-Based Contextual Answer Evaluation System. By integrating React, FastAPI, PostgreSQL, Sentence-BERT, KeyBERT, Machine Learning models, and teacher review, the system provides a complete framework for contextual descriptive answer evaluation. The phased implementation ensures that each module is developed, tested, and integrated properly to achieve an efficient, transparent, and reliable evaluation process.

| When                  | User                        | Action        | Description                        | IP        |
|-----------------------|-----------------------------|---------------|------------------------------------|-----------|
| Oct 6, 2026, 11:08 AM | Meera Krishnan<br>Admin     | user.login    | Meera Krishnan signed in           | 127.0.0.1 |
| Oct 6, 2026, 11:07 AM | Dr. Anitha Raman<br>Teacher | user.login    | Dr. Anitha Raman signed in         | 127.0.0.1 |
| Oct 6, 2026, 11:06 AM | Meera Krishnan<br>Admin     | user.login    | Meera Krishnan signed in           | 127.0.0.1 |
| Oct 6, 2026, 11:06 AM | Shradha S<br>Student        | user.login    | Shradha S signed in                | 127.0.0.1 |
| Oct 6, 2026, 11:05 AM | Dr. Anitha Raman<br>Teacher | answer.review | Reviewed Q1 of attempt #5 (4/5)    | 127.0.0.1 |
| Oct 6, 2026, 11:04 AM | Dr. Anitha Raman<br>Teacher | user.login    | Dr. Anitha Raman signed in         | 127.0.0.1 |
| Oct 6, 2026, 11:02 AM | Shradha S<br>Student        | attempt.start | Started 'Operating Systems Quiz 1' | 127.0.0.1 |

*Fig 5.9 Admin audit logs*

## **CHAPTER 6**

### **CONCLUSION AND FUTURE WORK**

#### **6.1 CONCLUSION**

The AI-Based Contextual Answer Evaluation System is developed to address the challenges associated with the manual evaluation of descriptive answers. Traditional answer evaluation requires significant time and effort from teachers and may result in variations in scoring and delayed feedback. The proposed system provides an AI-assisted approach that supports the evaluation process while maintaining the teacher as the final decision-maker.

The system integrates Natural Language Processing, Semantic Similarity Analysis, Concept Extraction, Rubric-Based Evaluation, and Machine Learning to analyze student answers. Sentence-BERT is used to understand the contextual similarity between the student answer and the reference answer, while KeyBERT is used to identify important concepts. The evaluation features are further processed using Random Forest Regressor and XGBoost Regressor for score prediction.

An important feature of the proposed system is the teacher-in-the-loop approach. AI-generated scores and feedback are reviewed by teachers before the final results are published. Teachers can modify scores, edit feedback, and approve or reject evaluations. This ensures that Artificial Intelligence assists the evaluation process without completely replacing human academic judgment.

The system also provides rubric-wise evaluation, personalized feedback, result management, analytics, and audit trail functionality. These features improve transparency and help students understand their performance and areas requiring improvement.

Thus, the proposed system provides an efficient and structured approach for evaluating digital descriptive answers. By combining contextual analysis, Machine Learning, and teacher supervision, the AI-Based Contextual Answer Evaluation System aims to reduce evaluation workload, improve consistency, and provide meaningful feedback while ensuring that final academic decisions remain under human control.

## **6.2 FUTURE WORK**

The proposed system can be further enhanced to improve its functionality, scalability, and evaluation capabilities. Currently, the system focuses mainly on digital text-based descriptive answer evaluation. Several advanced technologies can be integrated in the future to extend the capabilities of the system.

### **6.2.1 Handwritten Answer Evaluation**

The future version of the system can support handwritten answer evaluation by integrating Optical Character Recognition (OCR) and handwriting recognition technologies. This would enable students to upload scanned answer sheets for automated processing.

### **6.2.2 LLM-Based Feedback**

Advanced Large Language Models (LLMs) can be integrated to generate more detailed and personalised feedback. This can help provide better explanations regarding missing concepts, incorrect information, and areas requiring improvement.

### **6.2.3 Rag-Based Evaluation**

Retrieval-Augmented Generation (RAG) can be implemented to retrieve relevant academic content from authorised knowledge sources. This can improve contextual understanding and provide additional support for evaluating complex answers.

### **6.2.4 Explainable AI**

Explainable Artificial Intelligence (XAI) techniques such as SHAP can be integrated to explain how different evaluation features contribute to the predicted score. This can improve transparency and increase teacher confidence in AI-generated results.

### **6.2.5 Continuous Model Improvement**

The Machine Learning models can be continuously improved using teacher-approved evaluation data. Over time, the system can learn from historical evaluations and improve its score prediction performance.

### **6.2.6 Scalable Deployment**

The system can be deployed using cloud infrastructure to support a larger number of users and assessments. Technologies such as Redis caching, Celery-based background processing, and cloud deployment can improve system performance and scalability.

### **6.2.7 Advanced Analytics**

Future versions can provide more advanced analytics for teachers and institutions. This can include student learning trends, concept-wise performance, class-level comparison, and predictive performance analysis.

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# Submission Summary

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Track 9: Digital Education and Assistive Learning Systems

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**Paper Title**

AI-Based Contextual Answer Evaluation System

**Abstract**

Descriptive exams assess a student's understanding in a more thorough way than objective formats, but manual evaluation of them is slow, varies from evaluator to evaluator, and leaves little room for individualised feedback. In this paper, we present an AI based system for evaluating answers in context: the teacher creates and makes available a digital exam including a number of descriptive questions, reference answers and marking rubrics, and the students take the exam and submit their typed responses online. Each answer submitted is then assessed using a contextual semantic analysis process which compares the student's response with the reference answer and the rubric at the level of meaning instead of relying on keyword matching, resulting in a predicted grade, a confidence score and personalised feedback at the level of concepts. Rather than making the AI-generated results public immediately, each evaluation is sent to a teacher review and approval dashboard, where the teacher can check the result, make any necessary edits or override the score before it is sent to the student. This ensures that the teacher remains the last person to have authority over the grading while at the same time removing the need for the teacher to repeatedly read through and mark each answer from scratch. We explain the system's architecture, the pipeline for contextual evaluation and rubric-based scoring, the teacher in-the-loop review process, and the continuous-learning approach by which teacher corrections enhance future predictions, and we also describe an evaluation method comparing the system with manual grading and keyword matching baseline approaches.

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**Authors****SHRADHA S** (Rajalakshmi Engineering College) <230701311@rajalakshmi.edu.in>

ROSHINI V (Rajalakshmi Engineering College) &lt;230701270@rajalakshmi.edu.in&gt;

SAKSHI P (Rajalakshmi Engineering College) &lt;230701279@rajalakshmi.edu.in&gt;

SANDHIYA P (Rajalakshmi Engineering College) <230701282@rajalakshmi.edu.in>

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# AI-Based Contextual Answer Evaluation System

## Mrs. S. Punitha

Assistant Professor (Supervisor)  
Computer Science and Engineering  
Rajalakshmi Engineering College  
Chennai, India  
s.punitha@rajalakshmi.edu.in

## Shradha S

Student  
Computer Science and Engineering  
Rajalakshmi Engineering College  
Chennai, India  
230701311@rajalakshmi.edu.in

## Roshini V

Student  
Computer Science and Engineering  
Rajalakshmi Engineering College  
Chennai, India  
230701270@rajalakshmi.edu.in

## Sakshi P

Student  
Computer Science and Engineering  
Rajalakshmi Engineering College  
Chennai, India  
230701279@rajalakshmi.edu.in

## Sandhiya P

Student  
Computer Science and Engineering  
Rajalakshmi Engineering College  
Chennai, India  
230701282@rajalakshmi.edu.in

**Abstract—** *Descriptive exams assess a student's understanding in a more thorough way than objective formats, but manual evaluation of them is slow, varies from evaluator to evaluator, and leaves little room for individualised feedback. In this paper, we present an AI-based system for evaluating answers in context: the teacher creates and makes available a digital exam including a number of descriptive questions, reference answers and marking rubrics, and the students take the exam and submit their typed responses online. Each answer submitted is then assessed using a contextual semantic analysis process which compares the student's response with the reference answer and the rubric at the level of meaning instead of relying on keyword matching, resulting in a predicted grade, a confidence score and personalised feedback at the level of concepts. Rather than making the AI-generated results public immediately, each evaluation is sent to a teacher review and approval dashboard, where the teacher can check the result, make any necessary edits or override the score before it is sent to the student. This ensures that the teacher remains the last person to have authority over the grading while at the same time removing the need for the teacher to repeatedly read through and mark each answer from scratch. We explain the system's architecture, the pipeline for contextual evaluation and rubric-based scoring, the teacher-in-the-loop review process, and the continuous-learning approach by which teacher corrections enhance future predictions, and we also describe an evaluation method comparing the system with manual grading and keyword-matching baseline approaches.*

**Index Terms** - Artificial Intelligence, Natural Language Processing, Contextual Answer Evaluation, Semantic Similarity, Rubric-Based Scoring, Teacher-in-the-Loop Assessment, Educational Technology.

## I. INTRODUCTION

Because of the increasing use of digital learning platforms, teachers are now able to carry out examinations, gather the responses and handle academic records completely online. Yet in most digital examination systems the processes of creating

exams, collecting the answers and assessing the answers are still seen as separate. A teacher may create and publish an exam in digital form and a student can submit a typed answer digitally, but the assessment of descriptive, free-text responses is still done by hand, off-line, and independently of the system which collected the answer.

To automate the assessment of descriptive answers is fundamentally more difficult than assessing objective questions since correctness involves a range of acceptable explanations which can use different words, structures and degrees of detail to express the same meaning. When a teacher looks at two answers that are written in very different ways, they have to decide whether each one covers the necessary concepts and they have to make this judgement consistently across dozens or even hundreds of scripts, frequently while under time pressure. This results in three common problems: a great deal of teacher time is required for the evaluation, the marking differs from one evaluator to another and also for the same evaluator over the course of a long session, and students are given a mark without receiving any explanation of what was missing from their answer.

Simple auto-grading methods based on keyword matching deal only with the surface appearance of an answer. Such methods give credit for the inclusion of certain words no matter whether those words are used properly, and they impose penalties on correct answers that use valid alternative phrasings. Missing is a stage of evaluation which understands an answer in a contextual way—being able to recognise that two differently worded explanations can both meet a rubric criterion, while a response that simply repeats the vocabulary from the rubric without showing any understanding should not.

The paper suggests an AI-based system for the evaluation of answers which regards the processes of setting up an exam, having students submit their answers digitally, and carrying out contextual AI evaluation as part of a single, integrated workflow, the teacher remaining as the last person to make a decision. In this system, the teacher prepares an exam made up of a number of descriptive questions, each one having a reference answer and a marking rubric. The students then log

on, take the exam, and submit their typed answers for each question. The system subsequently assesses each answer in the context of the reference answer and the rubric, forecasts a grade, and produces feedback indicating which concepts were included, only partially included, or were missing. All the results generated by the AI are kept for the teacher to examine before they are given to the student, thus the system serving as a aid to grading rather than taking the place of the teacher's judgement.

This study does not present a single new algorithm, but rather combines exam construction, the submission of answers in digital form, contextual semantic evaluation, rubric-aware scoring, feedback generation, and teacher approval into a single interconnected pipeline, incorporating a continuous-learning process that makes use of teacher corrections to enhance future evaluations. The system functions at three levels of intelligence: the first assesses the meaning contained in a student's answer; the second compares that meaning with the rubric in order to predict a fair mark; and the third uses teacher corrections to generate signals that improve the following evaluation cycle.

The rest of the paper is structured as follows. Section II looks at previous work in the areas of automated answer scoring, semantic similarity methods, and teacher-in-the-loop assessment, and sets the proposed system in relation to the gaps identified. Section III outlines the methodology and system architecture that has been proposed. Section IV details the evaluation methodology that is planned and gives illustrative results which are still subject to full empirical validation. Section V concludes the paper.

## **II. LITERATURE REVIEW**

Automated assessment of free-text answers has been studied under several names, including Automatic Short Answer Grading (ASAG) and Automated Essay Scoring (AES). Early methods relied mainly on textual statistics, while more recent methods have focused on embedded-based semantic comparison. This section reviews the main lines of that work and positions the proposed system against the gaps that remain.

### ***2.1 Keyword and Rule-Based Auto-Graders***

The earliest automated grader systems scored answers by matching keywords or regular-expression patterns against a model answer. Such systems are fast and easy to deploy, but brittle: they cannot distinguish between a keyword used correctly and the same keyword incorrectly or out-of-context, and penalize valid answers that use synonyms or restructured sentences instead of the exact expected phrasing.

### ***2.2 Semantic Similarity-Based Grading***

To overcome the limitations of such keyword matching, later systems represent both the student answer and the reference answer as vectors and score similarity using cosine distance. Previous approaches used TF-IDF- or Latent Semantic Analysis-based vectors; more recent approaches use contextual sentence embeddings, which capture meaning

rather than word overlap and are better able to recognize paraphrased but correct answers. However, similarity does not tell a teacher which specific rubric criteria were met, merely how close two pieces of text are to one another.

### ***2.3 Rubric-Aware and Criterion-Based Scoring***

A separate body of literature decomposes a model answer into discrete marking criteria and scores each criterion individually rather than producing a single overall similarity score. Criterion-level scoring is closer to how real human examiners grade, in that it allows partial credit to be awarded and makes the final mark explainable in terms of which parts of the rubric were satisfied. Few systems, however, combine criterion-level semantic matching with a trained score-prediction model that also reports a confidence level for its own prediction.

### ***2.4 Automated Feedback Generation***

Beyond merely producing a mark, some research has explored generating natural-language feedback that explains a score, typically by highlighting matched and unmatched concepts between the student response and the reference answer. This feedback is most useful to students when it is specific to their own answer rather than a generic explanation of the rubric, and when it distinguishes between concepts that are completely missing and concepts that are present but incompletely explained.

### ***2.5 Human-in-the-Loop and Teacher-Assisted Grading***

A recurring concern with fully automated grading is the reliability for high-stakes assessment. Several studies argue that AI-assisted grading tools are best employed as a decision-support layer for teachers, allowing the system to handle the first pass while a human evaluator verifies, edits or overrides the result before it is released. This teacher-in-the-loop pattern increases trust in the system and provides a natural source of corrected labels that can be used to retrain the underlying model over time.

### ***2.6 Digital Examination and Learning Management Platforms***

Learning management systems and online examination platforms have enabled it to be easy to publish exams, collect typed responses and manage results digitally. However, most such platforms either only support objective question types for auto-scoring, or treat descriptive-answer scoring as an external manual process that occurs outside the platform, with results re-entered afterwards. This separation means the benefits of digital collection are not carried through to evaluation.

### ***2.7 Plagiarism and Integrity Checking***

Tools such as similarity-based plagiarism checkers are commonly used alongside grading to flag copied content between submissions. These tools typically operate independently of the scoring pipeline, so a flagged similarity issue and a predicted mark are not evaluated together when a teacher reviews a submission.

## 2.8 Continuous Model Improvement from Educator Feedback

Wherever AI-assisted grading systems are used across multiple cycles of assessment, teacher corrections to the model-predicted scores represent a valuable labeled dataset. Periodically retraining the scoring model on this corrected data can increase agreement with human graders over time, but requires some form of versioning to ensure that a retrained model can be validated before it takes the place of another.

## 2.9 Research Gap

Across the reviewed systems, four recurring gaps can be identified:

Semantic understanding is often shallow: many auto-graders still rely on substantial keyword or surface-overlap rather than contextual meaning.

Scoring is rarely rubric-explainable: overall similarity scores do not tell a teacher which criteria were or were not met.

Evaluation and teacher oversight are disconnected: AI-predicted scores are frequently published directly or reviewed outside the platform, rather than through an integrated approval step.

Feedback is generic rather than answer-specific: students receive a mark without a concept-level explanation of their own answer's shortcomings.

## 2.10 Positioning of the Proposed Work

In light of these gaps, the suggested system is able to tie together exam authoring, digital answer collection, contextual semantic evaluation, rubric-explainable scoring, concept-level feedback generation and teacher approval into a unified workflow. The process is as follows: Teacher Creates Exam → Reference Answer & Rubric → Student Attends Exam → Digital Answer Submission → Preprocessing → Contextual Semantic Analysis → Rubric-Criterion Matching → Score Prediction with Confidence → Feedback Generation → Teacher Review → Approve/Override → Result Published → Student Access → Continuous Learning. The primary contribution is the ability to tie together these stages to form a connected system in which grading assistance, rubric explainability and teacher authority act as one unit.

## III. METHODOLOGY

The platform being proposed has a modular, service-based structure which separates the real-time aspect of the examination from the more demanding AI evaluation and model-improvement procedures, thereby ensuring that a heavy AI workload when evaluating does not cause delays for students who are taking the exam.

### 3.1 Architectural Split: Exam Interaction and Evaluation Environments

The system consists of two environments which work together. The Exam Interaction Environment is the live, student- and

teacher-oriented part of the platform and is responsible for exam creation, the delivery of questions, the timed submission of answers, and the teacher review dashboard. It is built with low latency and ensures a quick response, since students are interacting with it during the examination period.

The heavier computational tasks are managed by the Evaluation and Analytics Environment: this includes preprocessing the answers that have been submitted, generating contextual embeddings, matching the answers to the criteria in the rubric, running the score-prediction model, producing the feedback text, and periodically retraining the scoring model using corrections approved by teachers. Answers that have been submitted are passed from the Exam Interaction Environment into this environment to be evaluated asynchronously, which ensures that the process of submitting an answer remains quick even though the AI evaluation is taking place in the background.

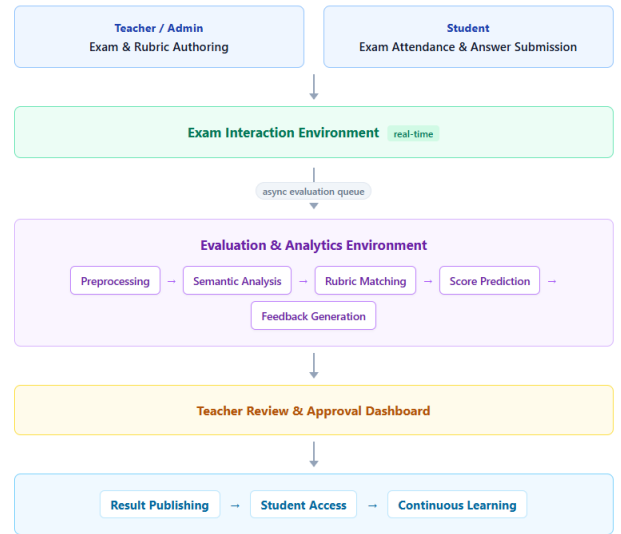


Figure 1: Two-Environment Architecture of the Proposed System — real-time exam interaction and asynchronous AI evaluation.

### 3.2 Data Ingestion and Feature Extraction Pipeline

The platform converts heterogeneous inputs — question papers, reference answers, rubrics and student answers — into a consistent feature structure before evaluation:

The text includes cleaned, tokenized and lemmatized forms of the student's answer and the reference answer, with stop words eliminated and the spelling made consistent.

The semantic features involve the creation of dense sentence-level embeddings for the student's answer, the reference answer, and each of the individual rubric criteria, and these are used to calculate similarity at the level of meaning rather than just by the amount of word overlap.

Concept Features: The key concepts and terms are extracted from both the student's answer and the reference answer, and

they are used to determine whether each rubric concept is present, partially present, or absent.

Features related to rubric alignment: instead of giving a single overall score, it provides a similarity score for the student's answer and each individual rubric component, so that partial credit can be explained.

In the case of historical features, when they were available, both the student's previous evaluated answers and the teacher's past correction patterns were used in order to identify any systematic bias between the teacher's tendency to grade and the AI's initial prediction.

### 3.3 Contextual Semantic Evaluation Core

The process used to predict a mark for an answer involves two stages. In the first stage, a contextual embedding model converts the student's answer, the reference answer, and each rubric criterion into dense vectors and then calculates the semantic similarity between the student's answer and each criterion separately. As a result, a rubric-aligned similarity profile is obtained instead of a single overall score. In the second stage, a trained ensemble scoring model uses this similarity profile along with features relating to concept overlap and answer length to predict a final mark together with a confidence score which shows how certain the model is of that prediction. Answers for which the prediction has low confidence are given priority for review by a teacher, so that the teacher's attention is focused on the answers about which the system is least certain rather than having to examine every submission to the same extent.

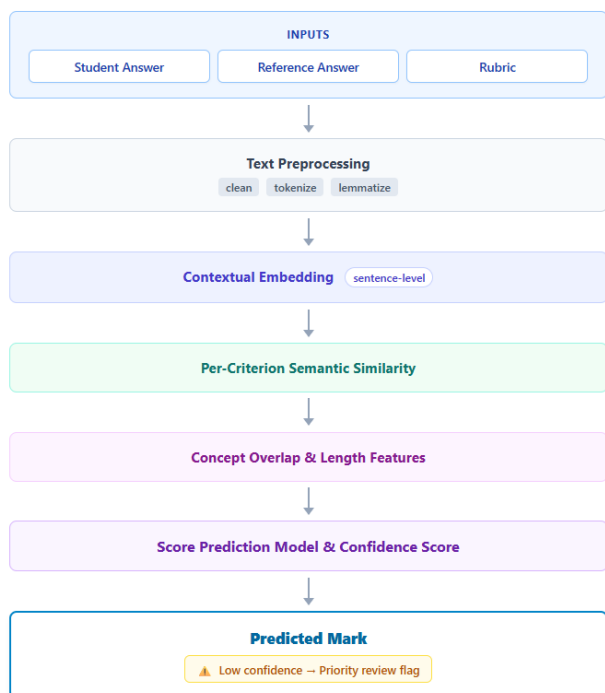


Figure 2: Contextual Semantic Evaluation and Rubric-Aligned Score Prediction Pipeline.

### 3.4 Feedback Generation Module

For every answer that is assessed, the system checks the concepts in the student's answer against those required by each rubric criterion and then produces a brief feedback summary which specifies the concepts that the student has explained correctly, the ones that were referred to but not fully explained, and the ones that were completely omitted. Since the feedback is based on the student's own answer and not on a general description of the rubric, students who have lost marks for different reasons receive different feedback.

### 3.5 Teacher Review, Approval and Continuous Learning

Instead of publishing the AI-generated marks directly, each evaluation—including the predicted mark, the per-criterion breakdown, the confidence score, and the generated feedback—is included in a teacher review dashboard. The teacher then has the option of accepting the result unchanged, modifying the mark or the feedback, or completely overriding it before approving it for release to the student. This procedure ensures that the teacher remains the final authority in grading. Each time a teacher makes a correction, this is recorded and then periodically used to retrain the score-prediction model, so that the system's predictions gradually come to match a particular teacher's or institution's grading habits over time. Before a newly retrained model replaces the one currently in use, it is validated using a separate set of evaluations that had previously been approved by the teacher, in order that retraining might not result in a silent deterioration of scoring quality.

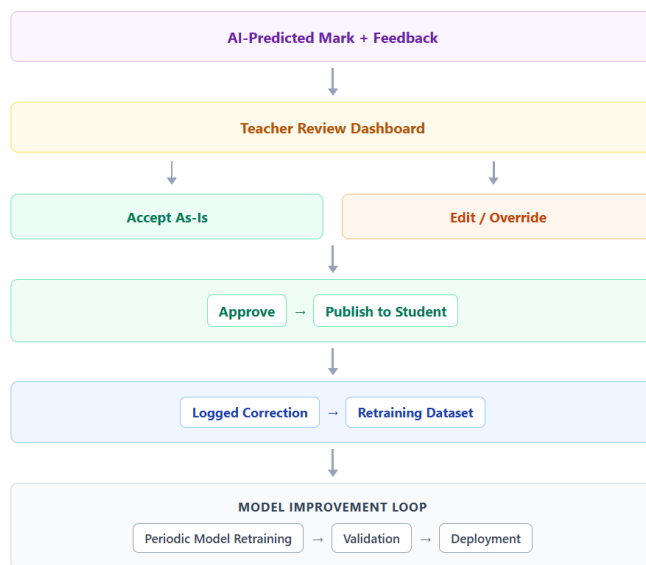


Figure 3: Teacher-in-the-Loop Review Workflow and Continuous-Learning Feedback Cycle.

## IV. EVALUATION METHODOLOGY AND RESULTS

This section describes planned evaluation methods and reports illustrative result structures. The figures below are placeholder values in the format the final evaluation will be presented in, to be replaced with measured results from testing against real

exam scripts and teacher grading once such testing has been completed.

#### 4.1 Planned Comparative Evaluation of Scoring Models

The proposed contextual pipeline (contextual embeddings with rubric-aligned ensemble scoring) will be compared against a keyword-matching baseline, a TF-IDF cosine-similarity baseline, and a single-model embedding-similarity baseline without rubric-criterion alignment. The primary evaluation metric is agreement with human (teacher) grading, measured using Quadratic Weighted Kappa (QWK) and Pearson correlation between predicted and teacher-assigned marks, alongside Mean Absolute Error (MAE) in mark points.

| Model / Approach                                  | QWK   | Pearson r | MAE (marks) |
|---------------------------------------------------|-------|-----------|-------------|
| Keyword Matching                                  | 0.184 | 0.482     | 2.183       |
| TF-IDF + Cosine Similarity                        | 0.078 | 0.379     | 3.101       |
| Single Embedding Similarity (no rubric alignment) | 0.249 | 0.276     | 0.941       |
| Proposed: Contextual + Rubric-Aligned Ensemble    | 0.379 | 0.461     | 0.736       |

Table 1: Planned Comparison of Scoring Approaches Against Human Grading.

#### 4.2 Teacher Override Rate and Evaluation Time Reduction

A second evaluation axis is the rate of teacher acceptance of AI-predicted marks without modification versus the rate of edits or overrides, and the amount of total grading time saved compared to fully manual evaluation of the same answer set. A decreasing override rate with successive cycles would indicate that the continuous-learning loop is improving agreement with a given teacher's grading tendency.

| Metric                                | Baseline (Manual) | With Proposed System* |
|---------------------------------------|-------------------|-----------------------|
| Avg. time per answer graded           | 3 min             | 1 min (reduced)       |
| Teacher override rate (initial cycle) | —                 | 30%                   |
| Time saved per 50-answer batch        | 125 min           | 50 min                |

Table 2: Planned Measurement of Teacher Workload and Override Rate.

#### 4.3 Feedback Quality Assessment

The generated feedback quality will be evaluated in two dimensions: concept-coverage accuracy (do the concepts flagged as missing or present actually match a teacher's independent assessment of the same answer) and perceived

usefulness, collected through a short student survey after the feedback has been released, following the approve-and-publish step described in Section 3.5.

#### 4.4 Low-Confidence Flagging Effectiveness

As the score-prediction model outputs a confidence value alongside each predicted mark, a planned analysis will investigate whether answers marked as low-confidence do in fact correspond to answers which teachers are more likely to edit or override, indicating that the confidence score is successfully directing the teacher's attention to the cases that need it most.

## V. CONCLUSION

This paper presented an AI-based contextual answer evaluation system which connects digital exam authoring, student answer submission, contextual semantic evaluation, rubric-explainable score prediction, concept-level feedback generation and teacher approval into a single workflow. Unlike keyword-matching auto-graders, it evaluates answers at the meaning level against individual rubric criteria, and unlike fully automated grading tools, it maintains the teacher as the final authority by requiring a review-and-approval step, while using teacher corrections to improve future predictions. We outlined the system architecture and the planned evaluation methodology against manual grading and keyword/embedding baselines. Future work includes completing empirical evaluation on live examination data across multiple subjects, extending support to multilingual answers, and investigating how low-confidence flagging can best direct limited teacher review time.

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# PROJECT\_PAPER.pdf

*by* Jones Emma

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**Mrs. S. Punitha**

1 Assistant Professor (Supervisor)  
Computer Science and Engineering  
Rajalakshmi Engineering College  
Chennai, India  
s.punitha@rajalakshmi.edu.in

**Shradha S**

1 Student  
Computer Science and Engineering  
Rajalakshmi Engineering College  
Chennai, India  
230701311@rajalakshmi.edu.in

**Roshini V**

1 Student  
Computer Science and Engineering  
Rajalakshmi Engineering College  
Chennai, India  
230701270@rajalakshmi.edu.in

**Sakshi P**

1 Student  
Computer Science and Engineering  
Rajalakshmi Engineering College  
Chennai, India  
230701279@rajalakshmi.edu.in

**Sandhiya P**

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Computer Science and Engineering  
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Because of the increasing use of digital learning platforms, teachers are now able to carry out examinations, gather the responses and handle academic records completely online. Yet in most digital examination systems the processes of creating exams, collecting the answers and assessing the answers are

still seen as separate. A teacher may create and publish an exam in digital form and a student can submit a typed answer digitally, but the assessment of descriptive, free-text responses is still done by hand, off-line, and independently of the system which collected the answer.

To automate the assessment of descriptive answers is fundamentally more difficult than assessing objective questions since correctness involves a range of acceptable explanations which can use different words, structures and degrees of detail to express the same meaning. When a teacher looks at two answers that are written in very different ways, they have to decide whether each one covers the necessary concepts and they have to make this judgement consistently across dozens or even hundreds of scripts, frequently while under time pressure. This results in three common problems: a great deal of teacher time is required for the evaluation, the marking differs from one evaluator to another and also for the same evaluator over the course of a long session, and students are given a mark without receiving any explanation of what was missing from their answer.

Simple auto-grading methods based on keyword matching deal only with the surface appearance of an answer. Such methods give credit for the inclusion of certain words no matter whether those words are used properly, and they impose penalties on correct answers that use valid alternative phrasings. Missing is a stage of evaluation which understands an answer in a contextual way—being able to recognise that two differently worded explanations can both meet a rubric criterion, while a response that simply repeats the vocabulary from the rubric without showing any understanding should not.

The paper suggests an AI-based system for the evaluation of answers which regards the processes of setting up an exam, having students submit their answers digitally, and carrying out contextual AI evaluation as part of a single, integrated workflow, the teacher remaining as the last person to make a decision. In this system, the teacher prepares an exam made up of a number of descriptive questions, each one having a reference answer and a marking rubric. The students then log on, take the exam, and submit their typed answers for each

question. The system subsequently assesses each answer in the context of the reference answer and the rubric, forecasts a grade, and produces feedback indicating which concepts were included, only partially included, or were missing. All the results generated by the AI are kept for the teacher to examine before they are given to the student, thus the system serving as a aid to grading rather than taking the place of the teacher's judgement.

This study does not present a single new algorithm, but rather combines exam construction, the submission of answers in digital form, contextual semantic evaluation, rubric-aware scoring, feedback generation, and teacher approval into a single interconnected pipeline, incorporating a continuous-learning process that makes use of teacher corrections to enhance future evaluations. The system functions at three levels of intelligence: the first assesses the meaning contained in a student's answer; the second compares that meaning with the rubric in order to predict a fair mark; and the third uses teacher corrections to generate signals that improve the following evaluation cycle.

The rest of the paper is structured as follows. Section II looks at previous work in the areas of automated answer scoring, semantic similarity methods, and teacher-in-the-loop assessment, and sets the proposed system in relation to the gaps identified. Section III outlines the methodology and system architecture that has been proposed. Section IV details the evaluation methodology that is planned and gives illustrative results which are still subject to full empirical validation. Section V concludes the paper.

## II. LITERATURE REVIEW

Automated assessment of free-text answers has been studied under several names, including Automatic Short Answer Grading (ASAG) and Automated Essay Scoring (AES). Early methods relied mainly on textual statistics, while more recent methods have focused on embedded-based semantic comparison. This section reviews the main lines of that work and positions the proposed system against the gaps that remain.

### 2.1 Keyword and Rule-Based Auto-Graders

The earliest automated grader systems scored answers by matching keywords or regular-expression patterns against a model answer. Such systems are fast and easy to deploy, but brittle: they cannot distinguish between a keyword used correctly and the same keyword incorrectly or out-of-context, and penalize valid answers that use synonyms or restructured sentences instead of the exact expected phrasing.

### 2.2 Semantic Similarity-Based Grading

To overcome the limitations of such keyword matching, later systems represent both the student answer and the reference answer as vectors and score similarity using cosine distance. Previous approaches used TF-IDF or Latent Semantic Analysis-based vectors; more recent approaches use contextual sentence embeddings, which capture meaning rather than word overlap and are better able to recognize

paraphrased but correct answers. However, similarity does not tell a teacher which specific rubric criteria were met, merely how close two pieces of text are to one another.

### 2.3 Rubric-Aware and Criterion-Based Scoring

A separate body of literature decomposes a model answer into discrete marking criteria and scores each criterion individually rather than producing a single overall similarity score. Criterion-level scoring is closer to how real human examiners grade, in that it allows partial credit to be awarded and makes the final mark explainable in terms of which parts of the rubric were satisfied. Few systems, however, combine criterion-level semantic matching with a trained score-prediction model that also reports a confidence level for its own prediction.

### 2.4 Automated Feedback Generation

Beyond merely producing a mark, some research has explored generating natural-language feedback that explains a score, typically by highlighting matched and unmatched concepts between the student response and the reference answer. This feedback is most useful to students when it is specific to their own answer rather than a generic explanation of the rubric, and when it distinguishes between concepts that are completely missing and concepts that are present but incompletely explained.

### 2.5 Human-in-the-Loop and Teacher-Assisted Grading

A recurring concern with fully automated grading is the reliability for high-stakes assessment. Several studies argue that AI-assisted grading tools are best employed as a decision-support layer for teachers, allowing the system to handle the first pass while a human evaluator verifies, edits or overrides the result before it is released. This teacher-in-the-loop pattern increases trust in the system and provides a natural source of corrected labels that can be used to retrain the underlying model over time.

### 2.6 Digital Examination and Learning Management Platforms

Learning management systems and online examination platforms have enabled it to be easy to publish exams, collect typed responses and manage results digitally. However, most such platforms either only support objective question types for auto-scoring, or treat descriptive-answer scoring as an external manual process that occurs outside the platform, with results re-entered afterwards. This separation means the benefits of digital collection are not carried through to evaluation.

### 2.7 Plagiarism and Integrity Checking

Tools such as similarity-based plagiarism checkers are commonly used alongside grading to flag copied content between submissions. These tools typically operate independently of the scoring pipeline, so a flagged similarity issue and a predicted mark are not evaluated together when a teacher reviews a submission.

## 2.8 Continuous Model Improvement from Educator Feedback

Wherever AI-assisted grading systems are used across multiple cycles of assessment, teacher corrections to the model-predicted scores represent a valuable labeled dataset. Periodically retraining the scoring model on this corrected data can increase agreement with human graders over time, but requires some form of versioning to ensure that a retrained model can be validated before it takes the place of another.

## 2.9 Research Gap

Across the reviewed systems, four recurring gaps can be identified:

Semantic understanding is often shallow: many auto-graders still rely on substantial keyword or surface-overlap rather than contextual meaning.

Scoring is rarely rubric-explainable: overall similarity scores do not tell a teacher which criteria were or were not met.

Evaluation and teacher oversight are disconnected: AI-predicted scores are frequently published directly or reviewed outside the platform, rather than through an integrated approval step.

Feedback is generic rather than answer-specific: students receive a mark without a concept-level explanation of their own answer's shortcomings.

## 2.10 Positioning of the Proposed Work

In light of these gaps, the suggested system is able to tie together exam authoring, digital answer collection, contextual semantic evaluation, rubric-explainable scoring, concept-level feedback generation and teacher approval into a unified workflow. The process is as follows: Teacher Creates Exam → Reference Answer & Rubric → Student Attends Exam → Digital Answer Submission → Preprocessing → Contextual Semantic Analysis → Rubric-Criterion Matching → Score Prediction with Confidence → Feedback Generation → Teacher Review → Approve/Override → Result Published → Student Access → Continuous Learning. The primary contribution is the ability to tie together these stages to form a connected system in which grading assistance, rubric explainability and teacher authority act as one unit.

## III. METHODOLOGY

The platform being proposed has a modular, service-based structure which separates the real-time aspect of the examination from the more demanding AI evaluation and model-improvement procedures, thereby ensuring that a heavy AI workload when evaluating does not cause delays for students who are taking the exam.

### 3.1 Architectural Split: Exam Interaction and Evaluation Environments

The system consists of two environments which work together. The Exam Interaction Environment is the live, student- and

teacher-oriented part of the platform and is responsible for exam creation, the delivery of questions, the timed submission of answers, and the teacher review dashboard. It is built with low latency and ensures a quick response, since students are interacting with it during the examination period.

The heavier computational tasks are managed by the Evaluation and Analytics Environment: this includes preprocessing the answers that have been submitted, generating contextual embeddings, matching the answers to the criteria in the rubric, running the score-prediction model, producing the feedback text, and periodically retraining the scoring model using corrections approved by teachers. Answers that have been submitted are passed from the Exam Interaction Environment into this environment to be evaluated asynchronously, which ensures that the process of submitting an answer remains quick even though the AI evaluation is taking place in the background.

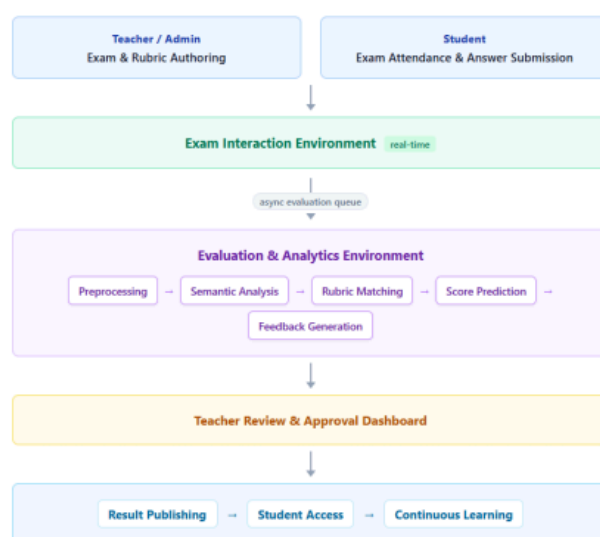


Figure 1: Two-Environment Architecture of the Proposed System — real-time exam interaction and asynchronous AI evaluation.

### 3.2 Data Ingestion and Feature Extraction Pipeline

The platform converts heterogeneous inputs — question papers, reference answers, rubrics and student answers — into a consistent feature structure before evaluation:

2 The text includes cleaned, tokenized and lemmatized forms of the student's answer and the reference answer, with stop words eliminated and the spelling made consistent.

The semantic features involve the creation of dense sentence-level embeddings for the student's answer, the reference answer, and each of the individual rubric criteria, and these are used to calculate similarity at the level of meaning rather than just by the amount of word overlap.

5 Concept Features: The key concepts and terms are extracted from both the student's answer and the reference answer, and

they **are** used to determine whether each rubric concept is present, partially present, or absent.

Features related to rubric alignment: instead of giving a single overall score, it provides a similarity score for the student's answer and each individual rubric component, so that partial credit can be explained.

In the case of historical features, when they were available, both the student's previous evaluated answers and the teacher's past correction patterns were used in order to identify any systematic bias between the teacher's tendency to grade and the AI's initial prediction.

3.3 Contextual Semantic Evaluation Core

The process used to predict a mark for an answer involves two stages. In the first stage, a contextual embedding model converts the student's answer, the reference answer, and each rubric criterion into dense vectors and then calculates the semantic similarity between the student's answer and each criterion separately. As a result, a rubric-aligned similarity profile is obtained instead of a single overall score. In the second stage, a trained ensemble scoring model uses this similarity profile along with features relating to concept overlap and answer length to predict a final mark together with a confidence score which shows how certain the model is of that prediction. Answers for which the prediction has low confidence are given priority for review by a teacher, so that the teacher's attention is focused on the answers about which the system is least certain rather than having to examine every submission to the same extent.

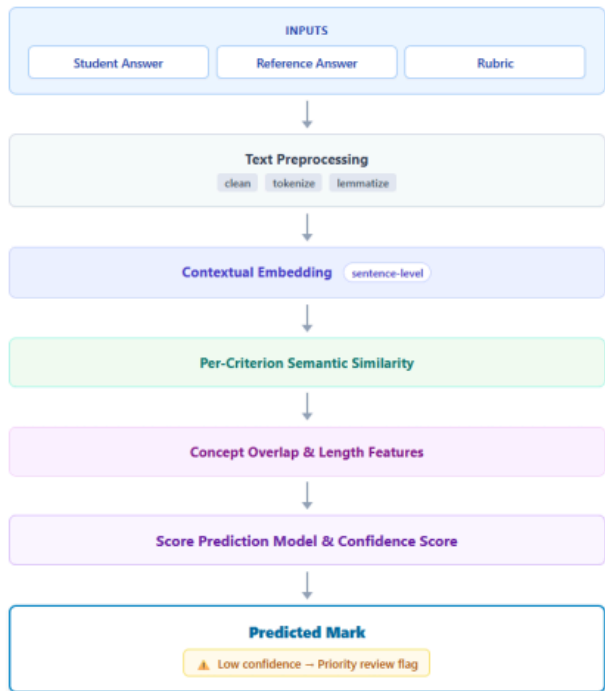


Figure 2: Contextual Semantic Evaluation and Rubric-Aligned Score Prediction Pipeline.

3.4 Feedback Generation Module

every answer that is assessed, the system checks the concepts in the student's answer against those required by each rubric criterion and then produces a brief feedback summary which specifies the concepts that the student has explained correctly, the ones that were referred to but not fully explained, and the ones that were completely omitted. Since the feedback is based on the student's own answer and not on a general description of the rubric, students who have lost marks for different reasons receive different feedback.

3.5 Teacher Review, Approval and Continuous Learning

Instead of publishing the AI-generated marks directly, each evaluation—including the predicted mark, the per-criterion breakdown, the confidence score, and the generated feedback—is included in a teacher review dashboard. The teacher then has the option of accepting the result unchanged, modifying the mark or the feedback, or completely overriding it before approving it for release to the student. This procedure ensures that the teacher remains the final authority in grading. Each time a teacher makes a correction, this is recorded and then periodically used to retrain the score-prediction model, so that the system's predictions gradually come to match a particular teacher's or institution's grading habits over time. Before a newly retrained model replaces the one currently in use, it is validated using a separate set of evaluations that had previously been approved by the teacher, in order that retraining might not result in a silent deterioration of scoring quality.

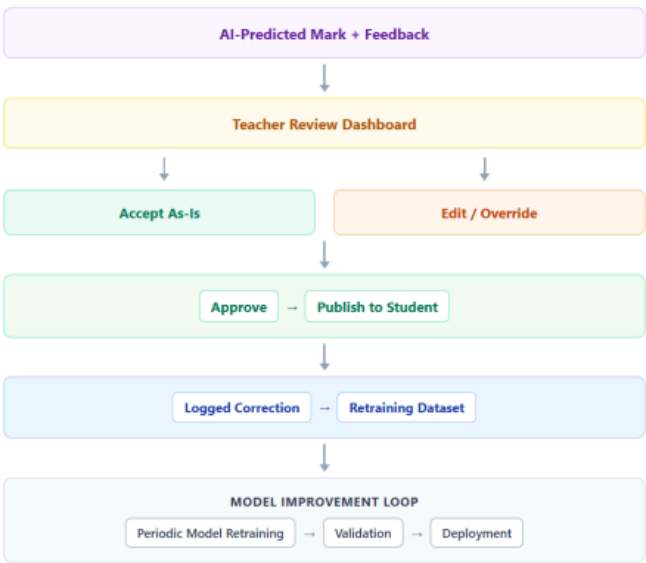


Figure 3: Teacher-in-the-Loop Review Workflow and Continuous Learning Feedback Cycle.

IV. EVALUATION METHODOLOGY AND RESULTS

This section describes planned evaluation methods and reports illustrative result structures. The figures below are placeholder values in the format the final evaluation will be presented in, to be replaced with measured results from testing against real

exam scripts and teacher grading once such testing has been completed.

#### 4.1 Planned Comparative Evaluation of Scoring Models

The proposed contextual pipeline (contextual embeddings with rubric-aligned ensemble scoring) will be compared against a keyword-matching baseline, a TF-IDF cosine-similarity baseline, and a single-model embedding-similarity baseline without rubric-criterion alignment. The primary evaluation metric is agreement with human (teacher) grading, measured using Quadratic Weighted Kappa (QWK) and Pearson correlation between predicted and teacher-assigned marks, alongside Mean Absolute Error (MAE) in mark points.

| Model / Approach                                  | QWK   | Pearson r | MAE (marks) |
|---------------------------------------------------|-------|-----------|-------------|
| Keyword Matching                                  | 0.184 | 0.482     | 2.183       |
| TF-IDF + Cosine Similarity                        | 0.078 | 0.379     | 3.101       |
| Single Embedding Similarity (no rubric alignment) | 0.249 | 0.276     | 0.941       |
| Proposed: Contextual + Rubric-Aligned Ensemble    | 0.379 | 0.461     | 0.736       |

Table 1: Planned Comparison of Scoring Approaches Against Human Grading.

#### 4.2 Teacher Override Rate and Evaluation Time Reduction

A second evaluation axis is the rate of teacher acceptance of AI-predicted marks without modification versus the rate of edits or overrides, and the amount of total grading time saved compared to fully manual evaluation of the same answer set. A decreasing override rate with successive cycles would indicate that the continuous-learning loop is improving agreement with a given teacher's grading tendency.

| Metric                                | Baseline (Manual) | With Proposed System* |
|---------------------------------------|-------------------|-----------------------|
| Avg. time per answer graded           | 3 min             | 1 min (reduced)       |
| Teacher override rate (initial cycle) | —                 | 30%                   |
| Time saved per 50-answer batch        | 125 min           | 50 min                |

Table 2: Planned Measurement of Teacher Workload and Override Rate.

#### 4.3 Feedback Quality Assessment

The generated feedback quality will be evaluated in two dimensions: concept-coverage accuracy (do the concepts flagged as missing or present actually match a teacher's independent assessment of the same answer) and perceived

usefulness, collected through a short student survey after the feedback has been released, following the approve-and-publish step described in Section 3.5.

#### 4.4 Low-Confidence Flagging Effectiveness

As the score-prediction model outputs a confidence value alongside each predicted mark, a planned analysis will investigate whether answers marked as low-confidence do in fact correspond to answers which teachers are more likely to edit or override, indicating that the confidence score is successfully directing the teacher's attention to the cases that need it most.

## V. CONCLUSION

This paper presented an AI-based contextual answer evaluation system which connects digital exam authoring, student answer submission, contextual semantic evaluation, rubric-explainable score prediction, concept-level feedback generation and teacher approval into a single workflow. Unlike keyword-matching auto-graders, it evaluates answers at the meaning level against individual rubric criteria, and unlike fully automated grading tools, it maintains the teacher as the final authority by requiring a review-and-approval step, while using teacher corrections to improve future predictions. We outlined the system architecture and the planned evaluation methodology against manual grading and keyword/embedding baselines. Future work includes completing empirical evaluation on live examination data across multiple subjects, extending support to multilingual answers, and investigating how low-confidence flagging can best direct limited teacher review time.

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